

Geospatial multi-criteria assessment of hydroelectric site suitability in the West African power pool

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Abstract

Hydropower is a vital renewable energy source for West Africa that addresses environmental concerns and energy security challenges. This study employed a geospatial multi-criteria decision analysis (MCDA) framework to assess hydroelectric site suitability across the West African Power Pool (WAPP). Nine biophysical parameters (drainage density, elevation, flow accumulation, geology, land cover, slope, soil texture, and stream power index) were sourced from authoritative datasets (SRTM DEM, CHIRPS, FAO, and UNESCO/ISRIC), processed, normalized, and integrated into a Composite Appropriateness Index (CAI) within a GIS environment. The results revealed spatial variability in hydroelectric potential, with the most suitable sites characterized by moderate drainage density, low to moderate elevation (below 536 m), and dendritic flow accumulation patterns. The geology is predominantly Precambrian, with land cover dominated by bare ground and rangeland, and rainfall intensity is higher in coastal regions. Slope gradients are generally low, indicating limited potential energy head for large-scale hydropower but suitability for low-head hydropower technologies. The soil textures are mainly loamy with clay and sand components, influencing infrastructure feasibility. Validation against existing hydropower infrastructure demonstrated strong spatial correspondence, confirming the model's effectiveness for site prioritization. While this study focuses on geospatial suitability mapping, it highlights the implications for hydropower development planning in WAPP and underscores the need for future integration of dynamic hydrological and energy modeling, as well as socioeconomic and infrastructural constraints to provide a comprehensive energy potential assessment. This study offers a valuable spatial decision-support tool to guide sustainable hydropower development and regional energy planning in West Africa.

Keywords

drainage density, elevation, flow accumulation, geology, hydropower, land cover and use, rainfall, soil texture, slope (tangents), stream power index, West Africa

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Introduction

This study aims to identify and map optimal sites for hydroelectric power generation within the West African Power Pool (WAPP) using a geospatial multi-criteria decision analysis (MCDA) framework. The persistent warming of the Earth's surface, driven by increased greenhouse gas emissions from human activities, has intensified climate-related challenges such as floods, droughts, and rising temperatures. As part of climate change mitigation strategies, renewable energy sources have become critical, with hydro energy emerging as a clean and abundant resource capable of addressing environmental concerns and growing energy security needs in West Africa.

Energy generation and distribution remain significant challenges across the African continent, impeding robust socioeconomic growth and development. Despite Africa's relatively low contribution to global greenhouse gas emissions, its energy sector accounts for a substantial share of these emissions. Sub-Saharan Africa struggles with limited power generation capacity and energy poverty, underscoring the need for sustainable and diversified energy solutions. The WAPP, comprising thirteen member nations, plays a strategic role in improving electricity access and reliability through regional cooperation, targeting ambitious goals of 100% electrification and 48% renewable energy by 2030.

Hydropower currently represents a vital energy source within the WAPP, yet a comprehensive understanding of the region's hydroelectric potential, considering geospatial and environmental factors, remains limited. This study addresses this gap by conducting a geospatial analysis of hydro energy site suitability across the WAPP region. The central research question guiding this work is: Which geospatial factors most significantly influence hydroelectric site suitability within the WAPP, and where are the optimal locations for hydropower development? We hypothesize that integrating parameters such as drainage density, elevation, flow accumulation, geology, land cover, rainfall, slope, soil texture, and stream power index through a Composite Appropriateness Index (CAI) will effectively delineate areas with high hydroelectric potential.

By employing GIS-based spatial analysis and an MCDA approach, this study systematically evaluates multiple biophysical parameters to quantify and map the spatial variability of hydroelectric suitability. The findings aim to provide decision-makers with a geospatial tool to prioritize hydroelectric development sites, support regional energy planning, and contribute to the sustainable development goals in West Africa. This study builds upon the existing literature by synthesizing diverse environmental datasets and applying a robust analytical framework tailored to the unique hydro-energy landscape of the WAPP region.

Literature review

Hydropower plays a critical role in addressing electrification challenges and promoting sustainable development across West Africa, particularly in the West African Power Pool (WAPP). The region's rural areas face significant energy deficits, and investment in renewable energy infrastructure, especially hydropower, offers a dual benefit: enhancing electricity access and supporting environmental sustainability through low greenhouse gas emissions. Therefore, a comprehensive assessment of hydro-energy potential that integrates hydrological, topographical, and socioeconomic factors is essential to optimize site selection and project viability within WAPP (ESMAP, 2023).

The hydrological regime, including river flow variability, rainfall-runoff dynamics, and evapotranspiration patterns, fundamentally determines hydropower potential across WAPP countries. Studies have revealed significant spatial variability in water balance components; for example, southern Senegal and Mali exhibit approximately 15% of rainfall generating runoff, while the central and northern regions demonstrate much lower runoff due to high evapotranspiration rates (Poyry Energy GmbH and ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), 2016a). Similar patterns are observed in Burkina Faso, Guinea, and The Gambia, where evapotranspiration often exceeds 80% of rainfall, limiting surface water availability for hydropower (Adama Nombé, Complementarities and synergies with intermittent renewable energy, related issues - Burkina Faso cases studies, 2022)(Adama Nombé, 2022); (THE GAMBIA). To capture this variability and support site viability assessment, hydrological modeling approaches such as the HBV-light model, ARIMA time series forecasting, and hydrological ensemble forecasting systems (HEFSs) have been effectively applied. For instance, the HBV-light model was calibrated for the Bafing River catchment to generate robust daily rainfall predictions (Didier Maria Ndione, 2020), while ARIMA-based perturbations improved rainfall ensemble forecasts (Energy, 2013). Moreover, the integration of GIS and remote sensing data further enhances spatial assessment of hydrological parameters, enabling detailed water balance analysis and runoff estimation critical for hydro-energy site evaluation (Poyry Energy GmbH and ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), 2016b).

Beyond hydrological and environmental factors, technical feasibility and socioeconomic considerations are vital in hydropower development. Key technical parameters such as flow rates, dam height, net head, and discharge capacity vary across micro and large-scale projects within WAPP. For example, the Souapiti dam in Guinea features a 130 m earth-fill dam with a 120 m head and 4500 m³/s spillway capacity (Amadou Sidibe, 2021; A. Dejoux, 2009), whereas micro-hydropower feasibility studies report total drop heights around 19 m with flow rates near 290 m³/s (Amadou Sidibe, 2021). Software tools like RETScreen

Table 1. Study area.

Ref	Study area	Type of site selection	Applied method	Evaluation criteria	Year
(Amédédjihundé H. J. Hounnou, 2019)	BENIN (The Yeripao site)	The run-of-river sizing	Unregulated Sorting Genetic Algorithm (NSGA-II).	The formal turbine flow rate (Q_{Tr}) and the hydropower units number (n_{hyd}).	2019
(Adama Nombré, Complementarities and synergies with intermittent renewable energy, related issues - Burkina Faso cases studies, 2022) (Poyry Energy GmbH and ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), 2016a) (M Moner-Girona, 2016)	Burkina Faso (Bagré & Bassiéri dams)	Hydroelectricity project sites, Basins.	30-limate model runs, the digital terrain model, VMAP0, SRTM30 and GIS base analysis.	Mean Annual Flow [m ³ /s], Forecast production [GWh/year], Mean Discharge [m ³ /s] and Topography	2016 2022 2016
(Mawusi Amenuvor, 2020) (Kwakwa, 2015)	Ghana (The Black Volta river)	Rainfall patterns	USGS provided Enhanced Thematic Mapper (ETM +), Multispectral Scanner (MSS), and Thematic Mapper (TM).	Hydrology, Evapotranspiration and Runoff [mm] and Elevation [m], rainfall patterns	2020
(Amadou Sidibe, 2021) (A. Dejoux, 2009) (Poyry Energy GmbH and ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), 2016b)	Guinea (The Niger, Milo and Corubal rivers)	Climate influence	RETSscreen Software, the region's meteorological station, topographic methodology, and the Global Positioning System (GPS).	Hydropower potential, Climate change	2021 2009 2016
(Raul., 2013), (Poyry Energy GmbH and ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), 2016c).	Guinea-Bissau (The Corubal River)	Hydropower production	General plant efficiency (turbines, hydraulic losses) is assumed 87.0% GIS base mapping	General information, Climate	2013 2016
(Salomon Obahoundje, 2023) (Claude Alain Kouadio, 2022)	Ivory Coast (The Bandama, Sassandra, White Bandama Watershed and Comoe rivers)	Overall suitability	Regional Climate Models of the CORDEX-Africa, QGIS and SWAT	Hydropower potential, Climate change, coefficient of determination R ² , p-factor, r-factor, and Nash–Sutcliffe (NS) efficiency coefficient.	2022 2022
(Alexandros Korkovelos, 2018) (Moussa Kanté, 2022)	Mali (The Niger, Bagoe and Senegal rivers)	Geospatial suitability assessment	Geographical Information Systems (GIS) and modern distant sensing, MESSAGE.	Mean Annual Flow [m ³ /s], Climate zones, Evapotranspiration and Runoff [mm]	2018 2022
(Prime Minister's Office, 2008) (Merem, 2006) (Energy, 2023)	Niger (The Niger River)	Hydro-Electro distribution	- Theoretical assessment - GIS mapping analysis	Hydrology, Hydropower potential and Electricity distribution	2008 2006 2023
(ABDULKADIR, 2013) (Adelakun, 2020) (Olayinka S. Ohunakin, 2011)	Nigeria (Kainji and Jebba dams)	hydropower reservoirs	The Gumbel (EVI) probability distribution model, the log-Pearson type III (LP3) model,	General information and environmental considerations,	2013 2020 2011

(continued)

Table 1. Continued.

Ref	Study area	Type of site selection	Applied method	Evaluation criteria	Year
(THE GAMBIA) (Poyry Energy GmbH and ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), 2016d) (The Gambia) (Faye, 2023) (Luciano Raso, 2018)	The Gambia (The Gambia River)	Hydro energy suitability	multilayer perceptron neural network, • GIS mapping analysis • CORDEX-Africa approach	coefficient of correlation, Attainable Hydro-Energy density and Climate considerations	2020 2016
(Taofic Bacharou, 2021) (Poyry Energy GmbH and ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), 2016e)	Senegal (Manantali dam)	River Basin	The COSMO-7 deterministic model and COSMO-LEPS probabilistic model, coupled with the PREVAH model, Stochastic Dual Dynamic Programming method.	Mean Yearly Release [m^3/s] and Hydropower Capability	2023 2018
(Alhaji Brima Gogra, 2010) (IRENA, 2023) (Poyry Energy GmbH and ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), 2016f)	Togo (The Mono River)	Topography assessment	• RCP4.5 and RCP8.5 • GIS mapping analysis	Annual and seasonal variations in discharge and elevation	2021 2016
	Sierra Leone	Future patterns Projections and statically based analysis	• RCP4.5 and RCP8.5 • GIS-based approach with statistical modeling	Climate zones, Hydropower Potential and future precipitation	2013 2023 2016

facilitate the sizing and structural design process by integrating meteorological, geological, and topographical data, supporting comprehensive feasibility analyses (Amadou Sidibe, 2021). Socioeconomic impacts and energy contributions of hydropower infrastructure are significant within WAPP; existing facilities such as the Manantali dam supply approximately 12% of Senegal's electricity (Didier Maria Ndione, 2020), while Ghana's Akosombo and Kpong dams collectively produce about 1060 MW, fulfilling 80% of national demand (Mawusi Amenuvor, 2020). These infrastructures underpin regional energy security and demonstrate the importance of integrating technical design with socioeconomic considerations in planning.

WAPP countries exhibit diverse hydropower capacities reflecting their geographical and climatic variability. Aggregate installed capacities include approximately 1988 MW in Nigeria, 1180 MW in Ghana, 605 MW in Côte d'Ivoire, and smaller but growing capacities in countries like Guinea (127 MW), Mali (155 MW), and Burkina

Faso (29 MW) (Energy, 2013) (Poyry Energy GmbH and ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), 2016c).

Many countries are undertaking expansions or renovations of existing stations, yet data gaps and inconsistencies remain, particularly in small-scale and mini-hydropower sectors. Consequently, the integration of geospatial and hydrological data is imperative to address these gaps and optimize hydro-energy resource allocation across the region (Energy, 2013) (Poyry Energy GmbH and ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), 2016d).

The literature highlights a variety of methodological frameworks that inform hydro-energy suitability assessments. Hydrological models such as HBV-light, coupled with statistical forecasting methods including ARIMA and kernel dressing techniques, improve the accuracy of rainfall and runoff predictions (Didier Maria Ndione, 2020); (Energy, 2013)). Hydrological ensemble forecasting systems provide probabilistic inputs that enhance decision-

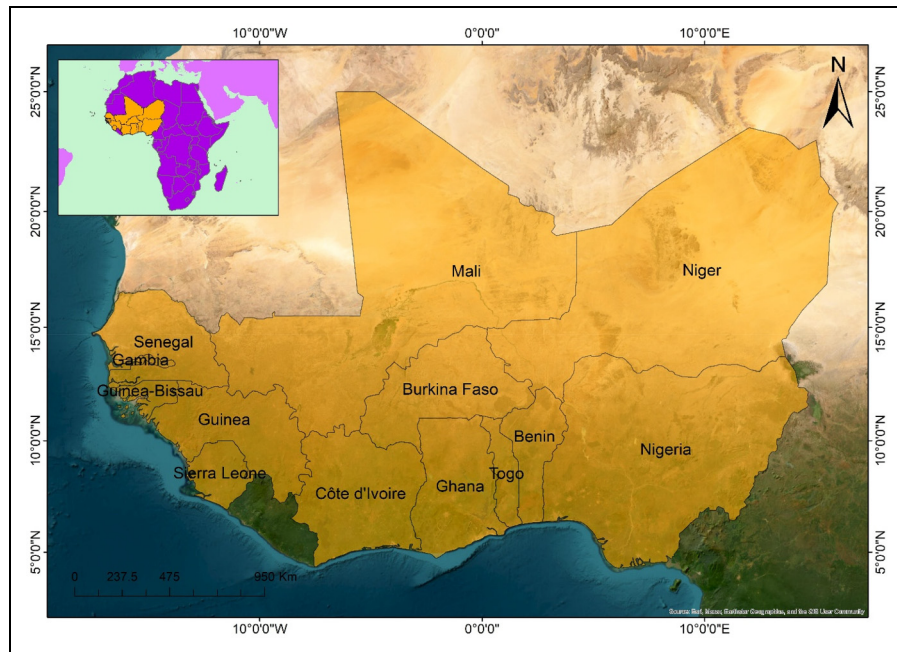


Figure 1. Study area map.

making under uncertainty ((Energy, 2013) GIS and remote sensing technologies enable high-resolution spatial analysis of hydrological parameters and terrain features, facilitating

Table 2. Key parameters and their data sources.

Parameter	Data source	Reference number in manuscript
Elevation, Drainage Density, Flow Accumulation, Slope (Tangent)	Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM) data	(Laboratory, Shuttle Radar Topography Mission (SRTM) Digital Elevation Model, 2013a) (Laboratory, SRTM – Elevation Data, 2013b)
Land Cover and Land Use	ESRI Living Atlas	(Esri, Land Use and Land Cover, 2020b)
Mean Annual Rainfall	Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS)	(Climate Hazards Group (CHG), 2015)
Soil Texture and Geology	Food and Agriculture Organization (FAO) and UNESCO/ISRIC databases	((FAO)), (Information, 2017)

All datasets were preprocessed to ensure spatial alignment, a uniform coordinate reference system, and consistent spatial resolution suitable for GIS analysis.

comprehensive geospatial evaluations of hydropower potential ((Energy, 2013) (Poyry Energy GmbH and ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), 2016e). These integrated approaches support the analytical workflow of hydro-energy suitability by enabling multi-criteria evaluation of site viability, incorporating hydrological variability, topographical constraints, and socioeconomic factors. Enhancements such as ensemble forecast quality improvements and spatial data integration underpin robust and scalable assessment frameworks suitable for WAPP's diverse contexts (Didier Maria Ndione, 2020; Poyry Energy GmbH and ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), 2016f).

In summary, the reviewed literature underscores the complexity and regional variability of hydropower potential within WAPP, shaped by hydrological regimes, environmental conditions, technical parameters, and

Table 3. Normalized weights.

Parameter	Weight (%)
Drainage Density	15
Elevation	15
Flow Accumulation	15
Geology	10
Land Cover	10
Rainfall	10
Slope (Tangent)	10
Soil Texture	7
Stream Power Index	8

Table 4. Overall hydroelectric suitability assessment.

Country	Low / very low (%)	Moderate (%)	High / very high (%)
Sierra Leone	3.68	42.95	53.37
Nigeria	40.50	45.36	14.13
Ghana	30.80	55.37	13.82
Guinea	14.23	75.95	9.82
Guinea Bissau			
Guinea	38.28	56.73	4.98
Togo	30.30	65.15	4.55
Benin	44.84	53.49	1.67
Cote d'Ivoire	47.87	46.67	5.45
Mali	73.39	23.83	2.78
Gambia	71.28	26.99	1.73
Niger	73.53	25.34	1.13
Burkina Faso	72.15	27.68	0.17
Senegal	68.52	31.38	0.09

socioeconomic contexts. While numerous studies provide valuable hydrological and infrastructural data, there remains a need for updated, harmonized datasets and

integrated modeling frameworks that combine hydrological, geospatial, and socioeconomic dimensions. This synthesis informs the current study's approach to evaluating hydro-energy suitability, emphasizing comprehensive, multi-faceted analyses to optimize hydropower development and support sustainable electrification goals across West Africa (ESMAP, 2023; Poyry Energy GmbH and ECOWAS Centre for Renewable Energy and Energy Efficiency (ECREEE), 2016g) Table 1.

Methodology

The study area

The study focuses on the West African Power Pool (WAPP) region, encompassing thirteen member countries: Benin, Burkina Faso, Ghana, Guinea, Guinea Bissau, Ivory Coast, Mali, Niger, Nigeria, The Gambia, Togo, Senegal, and Sierra Leone. The geographical extent spans latitudes 1.58°N to 25.00°N and longitudes 17.57°W to 15.95°E, covering diverse topographical, climatic, and environmental conditions relevant to hydroelectric potential assessment Figure 1.

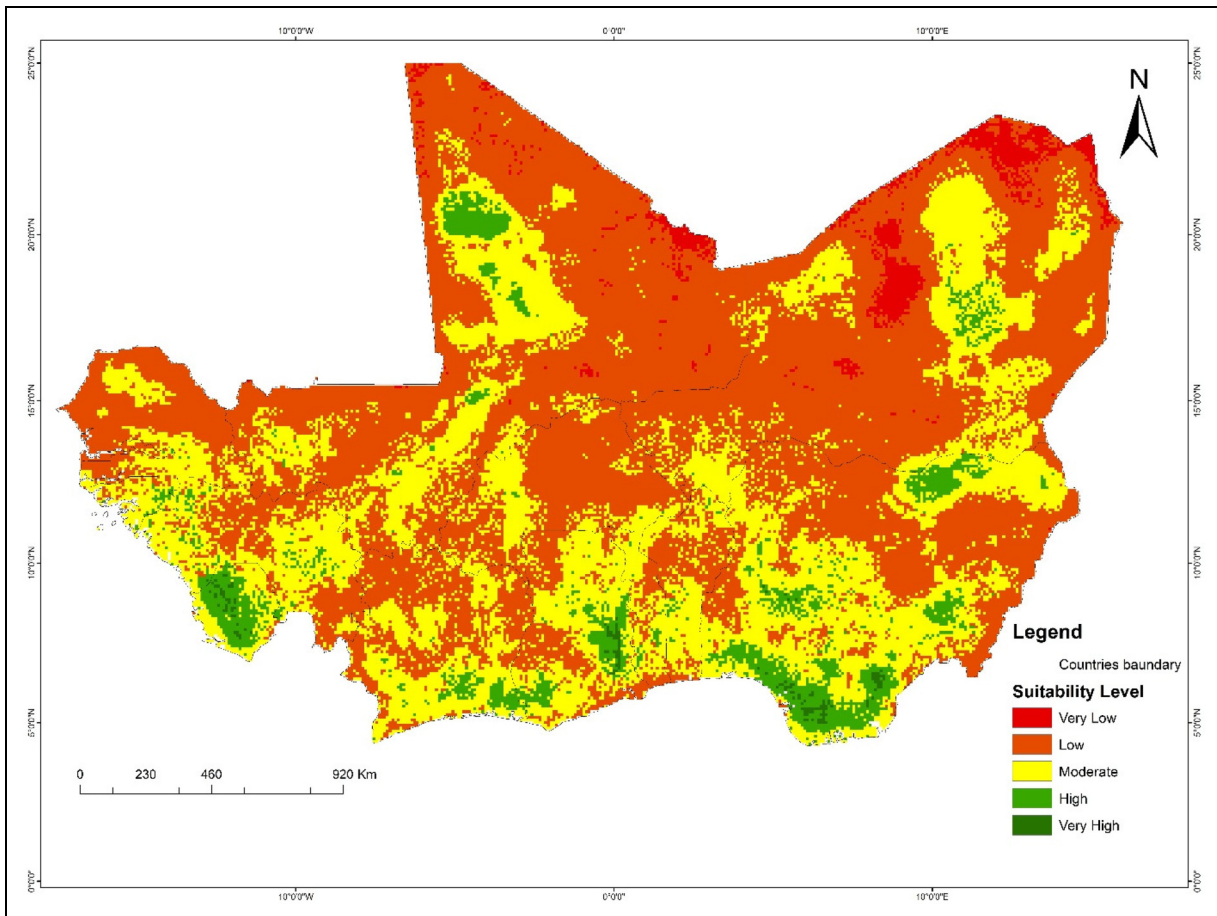
**Figure 2.** Overall Western suitability map.

Table 5. Western drainage density.

Mali			Niger		
Drainage Density (km/km ²)	Areas (km ²)	Percentage (%)	Drainage Density (km/km ²)	Areas (km ²)	Percentage (%)
0.0–0.025	173,603.00	12.62	0.0–0.025	130,431.89	10.04
0.025–0.035	495,038.70	35.98	0.025–0.035	329,512.61	25.36
0.035–0.046	426,536.88	31.00	0.035–0.046	420,638.56	32.37
0.046–0.061	232,573.52	16.90	0.046–0.061	329,212.73	25.34
0.061–0.097	48,267.44	3.51	0.061–0.097	89,511.15	6.89
	1,376,019.52	100.00		1,299,306.95	100.00
Burkina Faso			Benin		
Drainage Density (km/km ²)	Areas (km ²)	Percentage (%)	Drainage Density (km/km ²)	Areas (km ²)	Percentage (%)
0.0–0.025	32,877.32	11.52	0.0–0.025	8752.32	7.32
0.025–0.035	167,649.24	58.77	0.025–0.035	58,037.29	48.55
0.035–0.046	71,045.36	24.90	0.035–0.046	49,789.05	41.65
0.046–0.061	13,709.67	4.81	0.046–0.061	2953.37	2.47
0.061–0.097		0.00	0.061–0.097		0.00
	285,281.60	100.00		119,532.03	100.00
Nigeria			Togo		
Drainage Density (km/km ²)	Areas (km ²)	Percentage (%)	Drainage Density (km/km ²)	Areas (km ²)	Percentage (%)
0.0–0.025	110,023.29	11.80	0.0–0.025	6319.49	10.82
0.025–0.035	371,272.71	39.82	0.025–0.035	24,282.16	41.56
0.035–0.046	333,235.33	35.74	0.035–0.046	27,633.40	47.30
0.046–0.061	92,251.35	9.89	0.046–0.061	186.55	0.32
0.061–0.097	25,644.56	2.75	0.061–0.097		0.00
	932,427.23	100.00		58,421.60	100.00
Ghana			Cote d'Ivoire		
Drainage Density (km/km ²)	Areas (km ²)	Percentage (%)	Drainage Density (km/km ²)	Areas (km ²)	Percentage (%)
0.0–0.025	34,405.93	14.13	0.0–0.025	34,131.24	10.48
0.025–0.035	99,574.45	40.90	0.025–0.035	209,544.36	64.32
0.035–0.046	98,194.80	40.34	0.035–0.046	82,121.71	25.21
0.046–0.061	11,255.93	4.62	0.046–0.061		0.00
0.061–0.097		0.00	0.061–0.097		0.00
	243,431.11	100.00		325,797.31	100.00
Guinea			Senegal		
Drainage Density (km/km ²)	Areas (km ²)	Percentage (%)	Drainage Density (km/km ²)	Areas (km ²)	Percentage (%)
0.0–0.025	43,800.81	17.34	0.0–0.025	57,980.63	27.96
0.025–0.035	129,610.13	51.32	0.025–0.035	93,905.48	45.29
0.035–0.046	78,457.10	31.07	0.035–0.046	55,180.18	26.61
0.046–0.061	661.95	0.26	0.046–0.061	277.89	0.13
0.061–0.097		0.00	0.061–0.097		0.00
	252,530.00	100.00		207,344.18	100.00
Sierra Leone			Gambia		
Drainage Density (km/km ²)	Areas (km ²)	Percentage (%)	Drainage Density (km/km ²)	Areas (km ²)	Percentage (%)
0.0–0.025	16,520.91	22.58	0.0–0.025	4004.46	36.81
0.025–0.035	28,976.80	39.60	0.025–0.035	3497.91	32.15
0.035–0.046	27,495.90	37.57	0.035–0.046	3377.02	31.04
0.046–0.061	185.14	0.25	0.046–0.061		0.00
0.061–0.097		0.00	0.061–0.097		0.00
	73,178.75	100.00		10,879.40	100.00
Guinea Bissau					
Drainage Density (km/km ²)	Areas (km ²)	Percentage (%)			
0.0–0.025	15,445.13	46.27			
0.025–0.035	11,718.23	35.10			
0.035–0.046	6220.59	18.63			
0.046–0.061		0.00			
0.061–0.097		0.00			
	33,383.94	100.00			

Table 6. West elevation.

Mali			Niger		
Elevation (m)	Areas (km ²)	Percentage (%)	Elevation (m)	Areas (km ²)	Percentage (%)
–24.0–191.0	42,838.30	3.11	–24.0–191.0	3529.66	0.27
191.0–363.0	1,076,506.18	78.06	191.0–363.0	295,818.81	22.74
363.0–536.0	219,971.06	15.95	363.0–536.0	733,379.92	56.38
536.0–818.0	39,536.18	2.87	536.0–818.0	216,020.49	16.61
818.0–2359.0	179.93	0.01	818.0–2359.0	52,116.29	4.01
	1,379,031.65	100.00		1,300,865.16	100.00
Burkina Faso			Benin		
Elevation (m)	Areas (km ²)	Percentage (%)	Elevation (m)	Areas (km ²)	Percentage (%)
–24.0–191.0	3317.33	1.16	–24.0–191.0	28,385.25	23.76
191.0–363.0	265,907.28	93.21	191.0–363.0	68,154.90	57.06
363.0–536.0	14,851.55	5.21	363.0–536.0	22,143.35	18.54
536.0–818.0	1205.45	0.42	536.0–818.0	762.06	0.64
	285,281.60	100.00		119,445.56	100.00
Nigeria			Togo		
Elevation (m)	Areas (km ²)	Percentage (%)	Elevation (m)	Areas (km ²)	Percentage (%)
–24.0–191.0	253,020.46	27.06	–24.0–191.0	23,755.43	40.69
191.0–363.0	343,230.88	36.70	191.0–363.0	22,969.61	39.34
363.0–536.0	214,847.27	22.97	363.0–536.0	7662.87	13.12
536.0–818.0	98,592.52	10.54	536.0–818.0	3856.55	6.61
818.0–2359.0	25,493.21	2.73	818.0–2359.0	143.73	0.25
	935,184.33	100.00		58,388.19	100.00
Ghana			Cote d'Ivoire		
Elevation (m)	Areas (km ²)	Percentage (%)	Elevation (m)	Areas (km ²)	Percentage (%)
–24.0–191.0	149,600.85	61.52	–24.0–191.0	87,570.45	26.80
191.0–363.0	85,576.40	35.19	191.0–363.0	176,789.02	54.10
363.0–536.0	6685.70	2.75	363.0–536.0	57,013.64	17.45
536.0–818.0	1325.51	0.55	536.0–818.0	4927.38	1.51
818.0–2359.0	4.38	0.00	818.0–2359.0	495.36	0.15
	243,192.84	100.00		326,795.86	100.00
Guinea			Senegal		
Elevation (m)	Areas (km ²)	Percentage (%)	Elevation (m)	Areas (km ²)	Percentage (%)
–24.0–191.0	39,959.42	15.80	–24.0–191.0	206,789.88	99.17
191.0–363.0	29,179.70	11.54	191.0–363.0	1468.93	0.70
363.0–536.0	114,425.63	45.25	363.0–536.0	259.48	0.12
536.0–818.0	54,925.79	21.72	536.0–818.0	1.15	0.00
818.0–2359.0	14,366.01	5.68		208,519.44	100.00
	252,856.55	100.00			
Sierra Leone			Gambia		
Elevation (m)	Areas (km ²)	Percentage (%)	Elevation (m)	Areas (km ²)	Percentage (%)
–24.0–191.0	44,425.02	60.45	–24.0–191.0	11,189.23	100.00
191.0–363.0	10,869.17	14.79		11,189.23	100.00
363.0–536.0	15,487.04	21.07			
536.0–818.0	2418.01	3.29			
818.0–2359.0	289.27	0.39			
	73,488.51	100.00			
Guinea Bissau					
Elevation (m)	Areas (km ²)	Percentage (%)			
–24.0–191.0	33,863.65	99.79			
191.0–363.0	71.72	0.21			
	33,935.37	100.00			

Table 7. West flow accumulation.

Mali			Niger		
Flow Accumulation Value/km	Areas (km ²)	Percentage (%)	Flow Accumulation Value/km	Areas (km ²)	Percentage (%)
High	5583.21	10.40	High	4152.14	7.46
Low	48,100.36	89.60	Low	51,470.35	92.54
	53,683.57	100.00		55,622.49	100.00
Burkina Faso			Benin		
Flow Accumulation Value/km	Areas (km ²)	Percentage (%)	Flow Accumulation Value/km	Areas (km ²)	Percentage (%)
High	733.74	7.80	High	146.95	3.71
Low	8675.92	92.20	Low	3816.62	96.29
	9409.66	100.00		3963.56	100.00
Nigeria			Togo		
Flow Accumulation Value/km	Areas (km ²)	Percentage (%)	Flow Accumulation Value/km	Areas (km ²)	Percentage (%)
High	3537.55	10.14	High		0.00
Low	31,365.91	89.86	Low	2185.79	100.00
	34,903.46	100.00		2185.79	100.00
Ghana			Cote d'Ivoire		
Flow Accumulation Value/km	Areas (km ²)	Percentage (%)	Flow Accumulation Value/km	Areas (km ²)	Percentage (%)
High	1122.38	13.24	High	905.47	8.15
Low	7353.91	86.76	Low	10,207.94	91.85
	8476.29	100.00		11,113.41	100.00
Guinea			Senegal		
Flow Accumulation Value/km	Areas (km ²)	Percentage (%)	Flow Accumulation Value/km	Areas (km ²)	Percentage (%)
High	196.89	2.38	High	85.00	1.31
Low	8069.77	97.62	Low	6391.19	98.69
	8266.66	100.00		6476.19	100.00
Sierra Leone			Gambia		
Flow Accumulation Value/km	Areas (km ²)	Percentage (%)	Flow Accumulation Value/km	Areas (km ²)	Percentage (%)
High	89.65	3.56	High	278.80	60.55
Low	2431.50	96.44	Low	181.61	39.45
	2521.15	100.00		460.42	100.00
Guinea Bissau					
Flow Accumulation Value/km	Areas (km ²)	Percentage (%)			
High		0.00			
Low	898.16	100.00			
	898.16	100.00			

Data collection and sources

Spatial datasets were sourced from authoritative repositories to ensure precision, validity, and reliability. Key parameters and their data sources include: [Table 2](#).

Parameter selection and justification

Nine biophysical parameters were selected based on their established influence on hydroelectric potential and supported by literature and expert consultation: drainage density, elevation, flow accumulation, geology, land cover, rainfall, slope (tangent), soil texture, and stream power index (SPI). Each parameter reflects a critical aspect of hydropower feasibility, such as runoff availability (drainage density), potential energy head (elevation and slope), geological stability (geology), environmental constraints (land cover), water availability (rainfall), soil properties

affecting infrastructure (soil texture), and erosive power of streams (SPI).

Data processing and normalization

To ensure comparability across parameters with different units and scales, all raster layers were normalized using min-max normalization:

$$n_i = \frac{x_i - x_{min}}{x_{max} - x_{min}}$$

Where:

n_i is the normalized value

x_i is the original parameter value

(x_{max}, x_{min}) are the minimum and maximum values of the parameter within the study area.

This scales each parameter to a 0–1 range, facilitating integration into the composite index.

Table 8. West geology.

Mali			Niger		
Geology	Areas (km ²)	Percentage (%)	Geology	Areas (km ²)	Percentage (%)
Carboniferous	51,336.83	8.59	Carboniferous	68,946.92	30.92
Cretaceous	71,455.63	11.95	Cretaceous	102,818.76	46.11
Cretaceous and Jurassic	915.81	0.15	Cretaceous and Jurassic	5038.67	2.26
Devonian	4567.30	0.76	Cretaceous through Carboniferous	29,713.28	13.32
Holocene	469,575.50	78.54	Devonian	16,491.69	7.40
Internal Water Body	2993.48	0.50	Holocene	185,050.67	82.98
Lower Cretaceous	47,609.34	7.96	Internal Water Body	3860.69	1.73
Mesozoic-Paleozoic extrusive and intrusive rock	8775.49	1.47	Lower Cretaceous	119,815.78	53.73
Mesozoic extrusive and intrusive rock	27,865.09	4.66	Mesozoic extrusive and intrusive rock	8154.02	3.66
Ordovician-Cambrian	32,855.27	5.50	Ordovician	390.11	0.17
Pleistocene	102,180.30	17.09	Ordovician-Cambrian	21,444.64	9.62
Precambrian (undivided)	418,346.54	69.98	Permian	2654.82	1.19
Quaternary (undivided)	16,442.58	2.75	Permian-Carboniferous	193.39	0.09
Silurian-Ordovician	962.65	0.16	Pleistocene	24,046.94	10.78
Tertiary	120,099.13	20.09	Precambrian (undivided)	126,956.03	56.93
Tertiary and Cretaceous	11,483.80	1.92	Quaternary extrusive and intrusive rocks	2116.98	0.95
	597,851.07	100.00	Quaternary (undivided)	26,676.60	11.96
			Silurian	4751.27	2.13
			Tertiary	83,770.08	37.56
			Tertiary extrusive and intrusive rock	328.29	0.15
			Triassic and Permian	4481.38	2.01
			Upper and Middle Devonian	14,612.15	6.55
				223,009.32	100.00
Burkina Faso			Benin		
Geology	Areas (km ²)	Percentage (%)	Geology	Areas (km ²)	Percentage (%)
Holocene	1147.48	0.40	Cenozoic	5021.37	25.82
Mesozoic extrusive and intrusive rock	1857.08	0.65	Cretaceous	2250.88	11.57
Precambrian (undivided)	276,428.36	96.28	Holocene	3219.49	16.55
Quaternary (undivided)	524.69	0.18	Internal Water Body	196.37	1.01
Tertiary	7158.73	2.49	Mesozoic and Paleozoic	8761.44	45.05
	287,116.34	100.00	Pleistocene	1631.75	8.39
			Precambrian (undivided)	9519.20	48.94
			Subsurface salt dome	419.37	2.16
			Tertiary	3534.63	18.17
				19,449.55	100.00
Nigeria			Togo		
Geology	Areas (km ²)	Percentage (%)	Geology	Areas (km ²)	Percentage (%)
Cenozoic	23,782.29	2.53	Cenozoic	1005.89	1.74
Cretaceous	154,524.48	16.42	Cretaceous	352.26	0.61
Holocene	52,904.04	5.62	Holocene	678.89	1.17
Internal Water Body	12,557.25	1.33	Internal Water Body	17.06	0.03
Lower Cretaceous	32,737.98	3.48	Precambrian (undivided)	55,595.04	96.08
Mesozoic and Paleozoic	219.29	0.02	Subsurface salt dome	213.12	0.37
Mesozoic extrusive and intrusive rock	6224.42	0.66	Tertiary	910.56	1.57
Pleistocene	12,592.65	1.34		57,862.25	99.63
Precambrian (undivided)	428,007.67	45.47			

(continued)

Table 8. Continued.

Mali			Niger		
Quaternary (undivided)	128,069.33	13.61			
Subsurface salt dome	4826.01	0.51			
Tertiary	71,012.60	7.54			
Tertiary and Cretaceous	3223.58	0.34			
Tertiary extrusive and intrusive rock	10,620.02	1.13			
	941,301.60	100.00		58,077.14	100.00
Ghana			Cote d'Ivoire		
Geology	Areas (km ²)	Percentage (%)	Geology	Areas (km ²)	Percentage (%)
Carboniferous and Devonian	86.94	0.18	Holocene	1688.22	0.52
Cenozoic	673.41	1.41	Internal Water Body	387.69	0.12
Holocene	2085.45	4.36	Mesozoic extrusive and intrusive rock	200.12	0.06
Internal Water Body	12,745.60	26.66	Precambrian (undivided)	321,413.95	99.22
Ordovician-Cambrian	32,207.64	67.38	Quaternary (undivided)	256.08	0.08
Precambrian (undivided)	19,667.47	41.15	Subsurface salt dome	575.35	0.18
Subsurface salt dome	148.85	0.31	Tertiary	4436.54	1.37
Tertiary	176.36	0.37		323,946.05	100.00
	47,799.03	100.00			
Guinea			Senegal		
Geology	Areas (km ²)	Percentage (%)	Geology	Areas (km ²)	Percentage (%)
Cambrian	5296.24	10.07	Cambrian	4312.93	12.48
Cretaceous	256.90	0.49	Cretaceous	293.87	0.85
Devonian	6571.15	12.49	Holocene	29,575.63	85.60
Holocene	5251.36	9.98	Internal Water Body	328.75	0.95
Mesozoic extrusive and intrusive rock	35,229.63	66.97	Mesozoic extrusive and intrusive rock	40.32	0.12
Ordovician	20,296.76	38.58	Ordovician	118.21	0.34
Precambrian (undivided)	16,680.09	31.71	Precambrian (undivided)	28,658.88	82.95
Silurian	13,259.56	25.21	Quaternary extrusive and intrusive rocks	54.89	0.16
Subsurface salt dome	829.99	1.58	Quaternary (undivided)	967.44	2.80
Tertiary	125.23	0.24	Subsurface salt dome	369.82	1.07
Upper and Middle Devonian	586.31	1.11	Tertiary	14,510.53	42.00
	52,605.27	100.00	Tertiary extrusive and intrusive rock	11.62	0.03
				34,551.50	100.00
Sierra Leone			Gambia		
Geology	Areas (km ²)	Percentage (%)	Geology	Areas (km ²)	Percentage (%)
Holocene	8389.41	11.54	Subsurface salt dome	345.14	3.07
Internal Water Body	360.24	0.50	Tertiary	7535.88	66.92
Mesozoic extrusive and intrusive rock	168.89	0.23	Holocene	2420.30	21.49
Paleozoic extrusive and intrusive rock	278.16	0.38	Quaternary (undivided)	959.24	8.52
Precambrian (undivided)	63,472.10	87.34		11,260.56	100.00
Subsurface salt dome	1303.92	1.79			
	72,668.80	100.00			
Guinea Bissau					
Geology	Areas (km ²)	Percentage (%)			
Cambrian	1861.58	5.45			
Cretaceous	7183.35	21.03			

(continued)

Table 8. Continued.

Mali		Niger
Devonian	1945.69	5.70
Holocene	1901.22	5.57
Mesozoic extrusive and intrusive rock	441.42	1.29
Ordovician	1435.98	4.20
Precambrian (undivided)	1406.20	4.12
Quaternary (undivided)	139.84	0.41
Silurian	2053.56	6.01
Subsurface salt dome	1093.19	3.20
Tertiary	8896.28	26.05
Upper and Middle Devonian	5795.82	16.97
	34,154.11	100.00

Weighting scheme and composite appropriateness Index (CAI)

Weights were assigned to each parameter to reflect their relative importance in influencing hydroelectric suitability. Weight derivation involved comprehensive literature review and expert elicitation tailored to the WAPP context. Table 3 summarizes the normalized weights:

The Composite Appropriateness Index (CAI) for each spatial unit was computed as a weighted sum of normalized parameters:

$$CAI = \sum_{i=1}^n (w_i \times n_i)$$

Where:

w_i = is the normalized weight for parameter (i) n_i = is the normalized score for parameter (i)

This aggregation produced a continuous suitability surface representing the hydroelectric potential.

Addressing multicollinearity and parameter redundancy

Pearson correlation analysis was conducted to detect potential multicollinearity among parameters, notably between elevation and slope. Parameters with high correlation coefficients (>0.7) were examined, and weights were adjusted to minimize redundancy while preserving their distinct contributions. For instance, elevation and slope were assigned to complementary weights to reflect their unique influences on potential energy and water flow velocity.

Sensitivity analysis

The sensitivity analysis ($\pm 10\%$ weight variation) revealed that the model is most sensitive to changes in Rainfall and Elevation weights. A 10% increase in the weight of Rainfall expanded the 'High' suitability class in coastal regions by approximately 4.2%, while a similar increase

in Elevation weight shifted suitability clusters toward the Guinea Highlands. Despite these fluctuations, the spatial core of 'High' suitability zones remained consistent across 92% of the study area, confirming that the CAI provides a stable prioritization framework regardless of minor weighting uncertainties.

GIS integration and suitability mapping

All normalized and weighted parameter layers were integrated within a GIS environment using raster calculator tools. The CAI raster was generated to produce a hydroelectric suitability map for the WAPP region. Suitability classes (e.g., high, moderate, low) were defined based on CAI value thresholds to facilitate interpretation and decision-making.

Validation

Suitability results were validated by spatially overlaying the CAI map with existing hydropower infrastructure locations across WAPP countries. Correspondence between high suitability zones and operational hydropower plants was analyzed to assess model accuracy. Additionally, comparisons were made with prior regional hydropower assessments to contextualize findings and identify discrepancies.

Results and discussion

Overall hydroelectric suitability assessment

The integration of nine biophysical parameters into the Composite Appropriateness Index provides a differentiated spatial assessment of hydroelectric potential across the WAPP region. Unlike individual parameters, the CAI identifies priority zones by balancing water availability, topographical relief, and geological stability. The results indicate that while most of the West African territory falls within "Low" to "Moderate" suitability classes, specific

Table 9. West land cover.

Mali			Niger		
Class name	Areas (km ²)	Percentage (%)	Class name	Areas (km ²)	Percentage (%)
Water	3380.77	0.24	Water	1611.82	0.12
Trees	33,881.10	2.44	Trees	2299.19	0.18
Flooded Vegetation	2231.97	0.16	Flooded Vegetation	848.01	0.06
Crops	11,962.76	0.86	Crops	6958.65	0.53
Built Area	2717.57	0.20	Built Area	1019.09	0.08
Bare Ground	525,051.69	37.84	Bare Ground	656,136.85	50.14
Snow/Ice		0.00	Snow/Ice		0.00
Clouds		0.00	Clouds		0.00
Rangeland	808,237.88	58.25	Rangeland	639,863.48	48.89
	1,387,463.75	100.00		1,308,737.09	100.00
Burkina Faso			Benin		
Class name	Areas (km ²)	Percentage (%)	Class name	Areas (km ²)	Percentage (%)
Water	836.63	0.29	Water	653.71	0.54
Trees	13,516.39	4.71	Trees	28,266.86	23.48
Flooded Vegetation	126.23	0.04	Flooded Vegetation	136.13	0.11
Crops	1614.00	0.56	Crops	4813.44	4.00
Built Area	2993.02	1.04	Built Area	2813.38	2.34
Bare Ground	298.56	0.10	Bare Ground	14.56	0.01
Snow/Ice		0.00	Snow/Ice		0.00
Clouds		0.00	Clouds	1.00	0.00
Rangeland	267,731.64	93.25	Rangeland	83,685.95	69.52
	287,116.48	100.00		120,385.03	100.00
Nigeria			Togo		
Class name	Areas (km ²)	Percentage (%)	Class name	Areas (km ²)	Percentage (%)
Water	9052.23	0.96	Water	318.29	0.54
Trees	200,032.79	21.24	Trees	15,583.53	26.50
Flooded Vegetation	7878.32	0.84	Flooded Vegetation	5.45	0.01
Crops	74,394.09	7.90	Crops	2406.09	4.09
Built Area	26,015.05	2.76	Built Area	1785.64	3.04
Bare Ground	1148.14	0.12	Bare Ground	3.71	0.01
Snow/Ice		0.00	Snow/Ice		0.00
Clouds	31.44	0.00	Clouds		0.00
Rangeland	623,112.09	66.17	Rangeland	38,706.31	65.82
	941,664.15	100.00		58,809.03	100.00
Ghana			Cote d'Ivoire		
Class name	Areas (km ²)	Percentage (%)	Class name	Areas (km ²)	Percentage (%)
Water	9309.74	3.79	Water	2578.44	0.78
Trees	106,188.37	43.27	Trees	225,717.96	68.55
Flooded Vegetation	683.80	0.28	Flooded Vegetation	276.33	0.08
Crops	8675.81	3.53	Crops	5584.34	1.70
Built Area	8296.14	3.38	Built Area	5064.79	1.51
Bare Ground	99.83	0.04	Bare Ground	16.21	0.00
Snow/Ice		0.00	Snow/Ice		0.00
Clouds	35.98	0.01	Clouds	79.29	0.00
Rangeland	112,139.17	45.69	Rangeland	89,976.00	27.32
	245,428.84	100.00		334,545.73	100.00
Guinea			Senegal		
Class name	Areas (km ²)	Percentage (%)	Class name	Areas (km ²)	Percentage (%)
Water	2092.00	0.82	Water	2439.96	1.16
Trees	137,652.34	53.99	Trees	24,776.87	11.80
Flooded Vegetation	1789.50	0.70	Flooded Vegetation	2769.23	1.32
Crops	2656.96	1.04	Crops	18,581.97	8.85
Built Area	2747.83	1.08	Built Area	2942.44	1.40
Bare Ground	57.97	0.02	Bare Ground	646.33	0.31
Snow/Ice		0.00	Snow/Ice		0.00
Clouds		0.00	Clouds		0.00

(continued)

Table 9. Continued.

Mali			Niger		
Rangeland	107,985.95	42.35	Rangeland	157,856.08	75.16
	254,982.55	100.00		210,012.89	100.00
Sierra Leone			Gambia		
Class name	Areas (km ²)	Percentage (%)	Class name	Areas (km ²)	Percentage (%)
Water	1055.62	1.42	Water	450.73	3.97
Trees	59,370.60	79.67	Trees	2514.64	22.15
Flooded Vegetation	1382.44	1.86	Flooded Vegetation	702.12	6.18
Crops	309.65	0.42	Crops	2681.55	23.62
Built Area	764.28	1.03	Built Area	641.31	5.65
Bare Ground	18.86	0.03	Bare Ground	30.48	0.27
Snow/Ice		0.00	Snow/Ice		0.00
Clouds		0.00	Clouds		0.00
Rangeland	11,615.11	15.59	Rangeland	4332.19	38.16
	74,516.55	100.00		11,353.01	100.00
Guinea Bissau					
Class name	Areas (km ²)	Percentage (%)			
Water	1627.55	4.67			
Trees	22,488.02	64.54			
Flooded Vegetation	2394.54	6.87			
Crops	524.70	1.51			
Built Area	504.19	1.45			
Bare Ground	63.83	0.18			
Snow/Ice		0.00			
Clouds		0.00			
Rangeland	7242.53	20.78			
	34,845.35	100.00			

coastal and highland regions exhibit significant potential for hydropower development.

The quantitative distribution of these suitability classes is summarized in [Table 4](#), which replaces the previous binary results summary. This table provides the discriminative data required to prioritize regional energy investments.

Regional suitability mapping

The spatial distribution of these findings is visualized in [Figure 2](#) (the Overall Western Suitability Map). The map reveals a clear latitudinal gradient: the northern Sahelian countries are dominated by “Low” suitability (exceeding 65% of their territory) due to limited rainfall and low slope ([NASA, 2025](#)). In contrast, a “suitability corridor” is evident along the southern coast, specifically in Sierra Leone and Guinea, where the intersection of high flow accumulation and moderate elevations creates optimal conditions for both large-scale and run-of-river technologies.

Spatial characterization of individual siting criteria

The regional hydroelectric suitability summarized in the previous sections is the result of the spatial intersection of nine distinct biophysical and environmental parameters. To understand the specific drivers of suitability within

each WAPP member nation, it is necessary to examine the individual distribution of these criteria.

Topographical parameters, specifically elevation and slope gradients derived from the Shuttle Radar Topography Mission, establish the physical constraints for potential energy head and determine the feasible hydro-power typology differentiating between high-head storage and low-head run-of-river systems ([Laboratory, SRTM – Elevation Data, 2013b](#); [NASA, 2025](#)). Hydrological abundance is captured through mean annual rainfall and drainage density, which reveal the water availability gradient from the humid coastal south to the arid Sahelian north ([GHG, 2015](#); [Eresanya et al., 2017](#)). Furthermore, the structural and environmental viability of hydroelectric sites is characterized by geology, soil texture, and land cover, which influence dam foundation safety, reservoir permeability, and potential land-use conflicts ([Esri, Land Use and Land Cover, 2020b](#)).

The following subsections ([Tables 5 to 14](#) and [Figures 6–113](#)) provide a detailed, country-level quantitative breakdown and spatial visualization for each of these nine parameters, highlighting the unique biophysical profile of each country in the WAPP region.

[Table 5](#) presents the drainage density distribution across the WAPP region, ranging from 0.0 to 0.097 km/km². Most drainage densities fall within the 0.0 to 0.046 km/km²

Table 10. Western rain fall.

Mali			Niger		
Rainfall (mm)	Areas (km ²)	Percentage (%)	Rainfall (mm)	Areas (km ²)	Percentage (%)
1.89–333.02	942,628.83	68.00	1.89–333.02	1,109,771.46	84.81
333.02–855.85	268,516.16	19.37	333.02–855.85	198,670.95	15.18
855.85–1430.97	174,683.02	12.60	855.85–1430.97	22.05	0.00
1430.97–2250.07	350.23	0.03	1430.97–2250.07		0.00
2250.07–4445.97		0.00	2250.07–4445.97		0.00
	1,386,178.23	100.00		1,308,464.46	100.00
Burkina Faso			Benin		
Rainfall (mm)	Areas (km ²)	Percentage (%)	Rainfall (mm)	Areas (km ²)	Percentage (%)
1.89–333.02	1251.24	0.44	1.89–333.02		0.00
333.02–855.85	166,220.98	57.89	333.02–855.85	5843.37	4.87
855.85–1430.97	119,643.91	41.67	855.85–1430.97	108,139.42	90.10
1430.97–2250.07		0.00	1430.97–2250.07	6039.12	5.03
2250.07–4445.97		0.00	2250.07–4445.97		0.00
	287,116.13	100.00		120,021.92	100.00
Nigeria			Togo		
Rainfall (mm)	Areas (km ²)	Percentage (%)	Rainfall (mm)	Areas (km ²)	Percentage (%)
1.89–333.02	5737.71	0.61	1.89–333.02		0.00
333.02–855.85	223,722.36	23.85	333.02–855.85	21.34	0.04
855.85–1430.97	420,760.79	44.85	855.85–1430.97	38,686.82	65.85
1430.97–2250.07	213,656.51	22.78	1430.97–2250.07	19,958.28	33.97
2250.07–4445.97	74,185.41	7.91	2250.07–4445.97	84.34	0.14
	938,062.79	100.00		58,750.78	100.00
Ghana			Cote d'Ivoire		
Rainfall (mm)	Areas (km ²)	Percentage (%)	Rainfall (mm)	Areas (km ²)	Percentage (%)
1.89–333.02		0.00	1.89–333.02		0.00
333.02–855.85	423.38	0.17	333.02–855.85		0.00
855.85–1430.97	154,349.01	63.14	855.85–1430.97	216,066.54	65.88
1430.97–2250.07	87,745.86	35.90	1430.97–2250.07	99,816.90	30.44
2250.07–4445.97	1924.24	0.79	2250.07–4445.97	12,064.05	3.68
	244,442.49	100.00		327,947.49	100.00
Guinea			Senegal		
Rainfall (mm)	Areas (km ²)	Percentage (%)	Rainfall (mm)	Areas (km ²)	Percentage (%)
1.89–333.02		0.00	1.89–333.02	14,909.53	7.14
333.02–855.85		0.00	333.02–855.85	138,452.02	66.31
855.85–1430.97	87,844.05	34.73	855.85–1430.97	51,076.99	24.46
1430.97–2250.07	125,987.79	49.81	1430.97–2250.07	4346.82	2.08
2250.07–4445.97	39,113.72	15.46	2250.07–4445.97		0.00
	252,945.57	100.00		208,785.36	100.00
Sierra Leone			Gambia		
Rainfall (mm)	Areas (km ²)	Percentage (%)	Rainfall (mm)	Areas (km ²)	Percentage (%)
1.89–333.02		0.00	1.89–333.02		0.00
333.02–855.85		0.00	333.02–855.85	7885.86	71.57
855.85–1430.97		0.00	855.85–1430.97	3133.00	28.43
1430.97–2250.07	10,190.72	14.01	1430.97–2250.07		0.00
2250.07–4445.97	62,572.46	85.99	2250.07–4445.97		0.00
	72,763.18	100.00		11,018.86	100.00
Guinea Bissau					
Rainfall (mm)	Areas (km ²)	Percentage (%)			
1.89–333.02		0.00			
333.02–855.85		0.00			
855.85–1430.97	6658.93	20.55			
1430.97–2250.07	21,420.51	66.10			
2250.07–4445.97	4327.23	13.35			
	32,406.66	100.00			

Table II. Western slope.

Mali			Niger		
Slope (in Degree)	Areas (km ²)	Percentage (%)	Slope (in Degree)	Areas (km ²)	Percentage (%)
0.0–0.49	1,169,728.53	85.02	0.0–0.49	1,090,762.37	83.99
0.49–1.79	182,521.44	13.27	0.49–1.79	178,389.62	13.74
1.79–4.29	19,741.61	1.43	1.79–4.29	22,820.56	1.76
4.29–8.59	3577.56	0.26	4.29–8.59	5305.81	0.41
8.59–25.47	332.10	0.02	8.59–25.47	1381.83	0.11
	1,375,901.24	100.00		1,298,660.19	100.00
Burkina Faso			Benin		
Slope (in Degree)	Areas (km ²)	Percentage (%)	Slope (in Degree)	Areas (km ²)	Percentage (%)
0.0–0.49	233,403.53	81.82	0.0–0.49	68,895.04	57.74
0.49–1.79	49,566.44	17.37	0.49–1.79	47,833.08	40.08
1.79–4.29	2245.22	0.79	1.79–4.29	2296.25	1.92
4.29–8.59	66.41	0.02	4.29–8.59	305.16	0.26
8.59–25.47		0.00	8.59–25.47		0.00
	285,281.60	100.00		119,329.54	100.00
Nigeria			Togo		
Slope (in Degree)	Areas (km ²)	Percentage (%)	Slope (in Degree)	Areas (km ²)	Percentage (%)
0.0–0.49	523,056.86	56.09	0.0–0.49	26,472.32	45.38
0.49–1.79	346,414.74	37.15	0.49–1.79	25,568.97	43.83
1.79–4.29	40,881.13	4.38	1.79–4.29	4406.80	7.55
4.29–8.59	15,534.52	1.67	4.29–8.59	1566.97	2.69
8.59–25.47	6691.49	0.72	8.59–25.47	324.87	0.56
	932,578.75	100.00		58,339.93	100.00
Ghana			Cote d'Ivoire		
Slope (in Degree)	Areas (km ²)	Percentage (%)	Slope (in Degree)	Areas (km ²)	Percentage (%)
0.0–0.49	130,967.25	53.99	0.0–0.49	130,967.25	53.99
0.49–1.79	97,764.76	40.30	0.49–1.79	97,764.76	40.30
1.79–4.29	9764.29	4.03	1.79–4.29	9764.29	4.03
4.29–8.59	3582.23	1.48	4.29–8.59	3582.23	1.48
8.59–25.47	486.73	0.20	8.59–25.47	486.73	0.20
	242,565.26	100.00		242,565.26	100.00
Guinea			Senegal		
Slope (in Degree)	Areas (km ²)	Percentage (%)	Slope (in Degree)	Areas (km ²)	Percentage (%)
0.0–0.49	84,046.67	33.36	0.0–0.49	184,525.44	89.09
0.49–1.79	101,884.50	40.44	0.49–1.79	21,514.98	10.39
1.79–4.29	45,458.64	18.04	1.79–4.29	820.01	0.40
4.29–8.59	16,854.93	6.69	4.29–8.59	247.41	0.12
8.59–25.47	3720.07	1.48	8.59–25.47	5.54	0.00
	251,964.81	100.00		207,113.37	100.00
Sierra Leone			Gambia		
Slope (in Degree)	Areas (km ²)	Percentage (%)	Slope (in Degree)	Areas (km ²)	Percentage (%)
0.0–0.49	40,080.80	55.21	0.0–0.49	8760.75	81.18
0.49–1.79	21,155.40	29.14	0.49–1.79	2031.07	18.82
1.79–4.29	8486.96	11.69	1.79–4.29		0.00
4.29–8.59	2417.49	3.33	4.29–8.59		0.00
8.59–25.47	457.99	0.63	8.59–25.47		0.00
	72,598.64	100.00		10,791.83	100.00
Guinea Bissau					
Slope (in Degree)	Areas (km ²)	Percentage (%)			
0.0–0.49	13,660,635.44	50.10			
0.49–1.79	12,847,578.78	47.11			
1.79–4.29	755,300.20	2.77			
4.29–8.59	5436.78	0.02			
8.59–25.47		0.00			
	27,268,951.20	100.00			

Table 12. Western soil texture.

Mali			Niger		
Soil Texture	Areas (km ²)	Percentage (%)	Soil Texture	Areas (km ²)	Percentage (%)
CLAY	21,014.82	2.39	CLAY	587.73	0.05
CLAY_LOAM	229,756.31	26.13	CLAY_LOAM	204,416.62	18.77
LOAM	482,465.69	54.88	LOAM	440,348.28	40.42
LOAMY_SAND	39,768.93	4.52	LOAMY_SAND	217,394.58	19.96
SAND	106,134.64	12.07	SAND	226,589.31	20.80
SANDY_CLAY_LOAM	184,093.35	20.94	SANDY_CLAY_LOAM	5239.84	0.48
SANDY_LOAM	313,226.38	35.63	SANDY_LOAM	207,197.63	19.02
SILTY_CLAY	236.01	0.03	UWB	3808.37	0.35
UWB	9410.81	1.07	WATER	3303.17	0.30
WATER	1513.95	0.17		1,089,336.52	100.00
	879,140.39	100.00			
Burkina Faso			Benin		
Soil Texture	Areas (km ²)	Percentage (%)	Soil Texture	Areas (km ²)	Percentage (%)
CLAY	22,038.96	26.38	CLAY	3137.89	2.61
CLAY_LOAM	7925.76	9.49	LOAM	12,118.23	10.09
LOAM	35,034.02	41.94	SANDY_CLAY	4.72	0.00
LOAMY_SAND	8674.02	10.38	SANDY_CLAY_LOAM	86,866.08	72.32
SAND	9856.17	11.80	SANDY_LOAM	17,990.89	14.98
SANDY_CLAY	2329.80	2.79		120,117.81	100.00
SANDY_CLAY_LOAM	111,126.01	13.32			
SANDY_LOAM	8999.76	10.77			
	83,528.93	100.00			
Nigeria			Togo		
Soil Texture	Areas (km ²)	Percentage (%)	Soil Texture	Areas (km ²)	Percentage (%)
CLAY	19,634.54	7.84	CLAY	1648.44	2.81
CLAY_LOAM	53,835.46	21.49	LOAM	20,328.22	34.60
LOAM	115,022.52	45.92	LOAMY_SAND	584.57	0.99
LOAMY_SAND	15,831.88	6.32	SANDY_CLAY_LOAM	27,532.38	46.86
SAND	46,183.61	18.44	SANDY_LOAM	8664.76	14.75
SANDY_CLAY	2833.10	1.13	WATER	53.45	0.09
SANDY_CLAY_LOAM	29,374.26	11.73		58,758.38	100.00
SANDY_LOAM	38,276.22	15.28			
WATER	9965.14	3.98			
	250,508.00	100.00			
Ghana			Cote d'Ivoire		
Soil Texture	Areas (km ²)	Percentage (%)	Soil Texture	Areas (km ²)	Percentage (%)
CLAY	10,726.63	5.02	CLAY	687.17	0.21
CLAY_LOAM	792.50	0.37	CLAY_LOAM	9052.93	2.77
LOAM	35,064.66	16.42	LOAM	42,080.34	12.89
LOAMY_SAND	15,116.98	7.08	SANDY_CLAY_LOAM	142,928.17	43.79
SANDY_CLAY_LOAM	151,808.11	71.10	SANDY_LOAM	131,625.54	40.33
SANDY_LOAM	18,753.78	8.78	WATER	2467.03	0.76
WATER	12,380.54	5.80		326,374.15	100.00
	213,508.88	100.00			
Guinea			Senegal		
Soil Texture	Areas (km ²)	Percentage (%)	Soil Texture	Areas (km ²)	Percentage (%)
CLAY	11,020.48	4.35	CLAY	6517.19	7.90
CLAY_LOAM	13,306.82	5.25	CLAY_LOAM	7323.70	8.88
LOAM	56,583.09	22.33	LOAM	29,734.70	36.06
SANDY_CLAY_LOAM	159,193.56	62.81	LOAMY_SAND	9465.63	11.48
SANDY_LOAM	13,340.50	5.26	SAND	29,408.92	35.67
	253,444.46	100.00	SANDY_CLAY_LOAM	40,011.00	48.53
			SANDY_LOAM	8652.12	10.49
			WATER	693.91	0.84
				82,450.14	100.00

(continued)

Table 12. Continued.

Mali			Niger		
Sierra Leone			Gambia		
Soil Texture	Areas (km ²)	Percentage (%)	Soil Texture	Areas (km ²)	Percentage (%)
CLAY	1293.93	1.77	CLAY	12.53	0.11
CLAY_LOAM	945.28	1.29	CLAY_LOAM	4530.07	41.13
LOAM	20,752.04	28.38	LOAM	902.47	8.19
SANDY_CLAY_LOAM	44,891.57	61.40	SANDY_CLAY_LOAM	3201.82	29.07
SANDY_LOAM	5235.31	7.16	SANDY_LOAM	2366.43	21.49
WATER	98.52	0.13	WATER	325.40	2.95
	73,118.13	100.00		11,013.32	100.00
Guinea Bissau					
Soil Texture	Areas (km ²)	Percentage (%)			
CLAY	0.72	0.00			
CLAY_LOAM	4386.63	13.27			
LOAM	6781.66	20.52			
SANDY_CLAY_LOAM	6135.05	18.56			
SANDY_LOAM	15,752.50	47.65			
	33,056.56	100.00			

range. Country-specific percentages of drainage density within this range are: Mali (79.60%), Niger (67.77%), Burkina Faso (95.19%), Benin (97.53%), Nigeria (87.36%), Togo (99.68%), Ghana (95.38%), Côte d'Ivoire (100.0%), Guinea (99.74%), Senegal (99.87%), Sierra Leone (99.75%), The Gambia (100.0%), and Guinea Bissau (100.0%). illustrates a spatial mosaic characterized by predominantly low drainage density zones across the WAPP. [Figure 3](#) presents the drainage density distribution across the WAPP.

[Table 6](#) presents the elevation distribution within the WAPP region, ranging from −24.0 m to 2359.0 m. The majority of elevations fall below 536.0 m, with country-specific percentages as follows: Mali (97.12%), Niger (79.38%), Burkina Faso (99.58%), Benin (99.36%), Nigeria (86.73%), Togo (93.14%), Ghana (99.45%), Côte d'Ivoire (98.34%), Guinea (72.60%), Senegal (100.0%), Sierra Leone (96.32%), The Gambia (100.0%), and Guinea Bissau (100.0%). [Figure 4](#) illustrates the spatial distribution of elevation categories, emphasizing the predominance of low-lying zones across the region.

[Table 7](#) summarizes flow accumulation values categorized as high or low across WAPP countries. Most countries exhibit a high percentage of low flow accumulation areas, including Mali (89.60%), Niger (92.54%), Burkina Faso (92.20%), Benin (96.29%), Nigeria (89.86%), Togo (100.0%), Ghana (86.67%), Côte d'Ivoire (91.85%), Guinea (97.62%), Senegal (98.69%), Sierra Leone (96.44%), and Guinea Bissau (100.0%). The Gambia stands out with 60.55% high flow accumulation. [Figure 5](#) depicts the dendritic flow accumulation pattern characteristic of the WAPP.

[Table 8](#) details the geological formations across WAPP countries, including Carboniferous, Cretaceous, Devonian, Holocene, Precambrian (undivided), and others. The most prevalent formation is Precambrian (undivided), dominating

most countries except The Gambia and Guinea Bissau, which have 4.12% Precambrian geology. Other significant formations include Holocene deposits in Mali (78.54%) and Niger (82.98%), and Mesozoic-Paleozoic rocks in Benin (45.05%). [Figure 6](#) shows the spatial distribution of these geological units across the region.

[Table 9](#) reports land cover classes within the WAPP, including water, trees, flooded vegetation, crops, built-up areas, bare ground, and rangeland. Bare ground and rangeland dominate most countries, such as Mali (37.84% bare ground, 58.25% rangeland) and Niger (50.14% bare ground, 48.89% rangeland). Tree cover is significant in countries like Benin (23.48%), Nigeria (21.24%), Ghana (43.27%), and Guinea (68.55%). [Figure 7](#) illustrates the spatial patterns of land cover types, highlighting the prevalence of bare ground and rangeland.

[Table 10](#) shows rainfall distribution ranging from 1.89 mm to 4445.97 mm annually. Most countries receive rainfall within the 1.89–1430.97 mm range, including Mali, Niger, Burkina Faso, and The Gambia at 100%. Higher rainfall (>1430.97 mm) is recorded predominantly in Sierra Leone (100%) and Guinea Bissau (79.45%). [Figure 8](#) maps rainfall intensity gradients, emphasizing higher precipitation along coastal areas compared to the drier northern regions.

[Table 11](#) presents slope classifications in degrees, with the majority of areas exhibiting low slope gradients between 0.0° and 1.79°. Country-specific percentages within this range include Mali (97.73%), Burkina Faso (99.19%), Benin (97.82%), Nigeria (83.24%), Sierra Leone (89.21%), The Gambia (100%), and Guinea Bissau (97.21%). [Figure 9](#) depicts the predominance of low slope gradients across the WAPP.

[Table 12](#) summarizes soil texture types, including clay, clay loam, loam, loamy sand, sand, sandy clay loam, and

Table 13. Western SPI (stream power Index).

Mali			Niger		
SPI (000 000 000)	Areas (km ²)	Percentage (%)	SPI (000 000 000)	Areas (km ²)	Percentage (%)
0.0–2.18	1,375,829.53	99.99	0.0–2.18	1,298,417.09	99.97
2.18–8.73	128.62	0.01	2.18–8.73	207.22	0.02
8.73–20.53	7.64	0.00	8.73–20.53	95.85	0.01
20.53–53.73		0.00	20.53–53.73	4.90	0.00
53.73–111.41		0.00	53.73–111.41		0.00
	1,375,965.79	100.00		1,298,725.05	100.00
Burkina Faso			Benin		
SPI (000 000 000)	Areas (km ²)	Percentage (%)	SPI (000 000 000)	Areas (km ²)	Percentage (%)
0.0–2.18	285,281.60	100.00	0.0–2.18	119,315.37	99.99
2.18–8.73		0.00	2.18–8.73	9.65	0.01
8.73–20.53		0.00	8.73–20.53	2.61	0.00
20.53–53.73		0.00	20.53–53.73	2.15	0.00
53.73–111.41		0.00	53.73–111.41		0.00
	285,281.60	100.00		119,329.78	100.00
Nigeria			Togo		
SPI (000 000 000)	Areas (km ²)	Percentage (%)	SPI (000 000 000)	Areas (km ²)	Percentage (%)
0.0–2.18	932,040.34	99.94	0.0–2.18	58,341.64	100.00
2.18–8.73	375.37	0.04	2.18–8.73		0.00
8.73–20.53	149.35	0.02	8.73–20.53		0.00
20.53–53.73	36.59	0.00	20.53–53.73		0.00
53.73–111.41	2.15	0.00	53.73–111.41		0.00
	932,603.80	100.00		58,341.64	100.00
Ghana			Cote d'Ivoire		
SPI (000 000 000)	Areas (km ²)	Percentage (%)	SPI (000 000 000)	Areas (km ²)	Percentage (%)
0.0–2.18	242,511.74	99.97	0.0–2.18	325,545.80	100.00
2.18–8.73	50.07	0.02	2.18–8.73		0.00
8.73–20.53	11.71	0.01	8.73–20.53		0.00
20.53–53.73	5.54	0.00	20.53–53.73		0.00
53.73–111.41		0.00	53.73–111.41		0.00
	242,579.05	100.00		325,545.80	100.00
Guinea			Senegal		
SPI (000 000 000)	Areas (km ²)	Percentage (%)	SPI (000 000 000)	Areas (km ²)	Percentage (%)
0.0–2.18	251,978.20	100.00	0.0–2.18	207,111.35	100.00
2.18–8.73		0.00	2.18–8.73		0.00
8.73–20.53		0.00	8.73–20.53		0.00
20.53–53.73		0.00	20.53–53.73		0.00
53.73–111.41		0.00	53.73–111.41		0.00
	251,978.20	100.00		207,111.35	100.00
Sierra Leone			Gambia		
SPI (000 000 000)	Areas (km ²)	Percentage (%)	SPI (000 000 000)	Areas (km ²)	Percentage (%)
0.0–2.18	72,612.47	100.00	0.0–2.18	10,799.98	100.00
2.18–8.73		0.00	2.18–8.73		0.00
8.73–20.53		0.00	8.73–20.53		0.00
20.53–53.73		0.00	20.53–53.73		0.00
53.73–111.41		0.00	53.73–111.41		0.00
	72,612.47	100.00		10,799.98	100.00
Guinea Bissau					
SPI (000 000 000)	Areas (km ²)	Percentage (%)			
0.0–2.18	32,746.16	100.00			
2.18–8.73		0.00			
8.73–20.53		0.00			
20.53–53.73		0.00			
53.73–111.41		0.00			
	32,746.16	100.00			

Table 14. Quantitative summary of hydroelectric suitability by country (% of national territory).

Mali			Niger		
Suitability level	Areas	Percentage	Suitability level	Areas	Percentage
Very Low	27,584.74	2.01	Very Low	95,754.31	7.40
Low	977,733.76	71.38	Low	856,134.07	66.13
Moderate	326,405.90	23.83	Moderate	328,093.97	25.34
High	38,028.06	2.78	High	14,690.60	1.13
Very High		0.00	Very High		0.00
	1,369,752.46	100.00		1,294,672.95	100.00
Burkina Faso			Benin		
Suitability level	Areas	Percentage	Suitability level	Areas	Percentage
Very Low		0.00	Very Low		0.00
Low	205,838.33	72.15	Low	53,449.63	44.84
Moderate	78,957.25	27.68	Moderate	63,766.65	53.49
High	486.02	0.17	High	1987.08	1.67
Very High		0.00	Very High		0.00
	285,281.60	100.00		119,203.36	100.00
Nigeria			Togo		
Suitability level	Areas	Percentage	Suitability level	Areas	Percentage
Very Low	91.00	0.01	Very Low		0.00
Low	375,515.17	40.49	Low	17,700.88	30.30
Moderate	420,686.16	45.36	Moderate	38,066.70	65.15
High	123,530.90	13.32	High	2658.03	4.55
Very High	7531.59	0.81	Very High		0.00
	927,354.82	100.00		58,425.61	100.00
Ghana			Code d'Ivoire		
Suitability level	Areas	Percentage	Suitability level	Areas	Percentage
Very Low		0.00	Very Low		0.00
Low	74,564.83	30.80	Low	154,864.82	47.87
Moderate	134,034.24	55.37	Moderate	150,977.19	46.67
High	29,727.87	12.28	High	17,646.20	5.45
Very High	3729.72	1.54	Very High		0.00
	242,056.66	100.00		323,488.21	100.00
Guinea			Senegal		
Suitability level	Areas	Percentage	Suitability level	Areas	Percentage
Very Low		0.00	Very Low		0.00
Low	95,743.91	38.28	Low	140,525.09	68.52
Moderate	141,867.57	56.73	Moderate	64,362.57	31.38
High	12,410.67	4.96	High	193.35	0.09
Very High	62.30	0.02	Very High		0.00
	250,084.45	100.00		205,081.02	100.00
Sierra Leone			Gambia		
Suitability level	Areas	Percentage	Suitability level	Areas	Percentage
Very Low		0.00	Very Low		0.00
Low	2600.88	3.68	Low	7194.81	71.28
Moderate	30,384.56	42.95	Moderate	2724.51	26.99
High	34,928.96	49.37	High	174.67	1.73
Very High	2830.62	4.00	Very High		0.00
	70,745.03	100.00		10,093.98	100.00
Guinea Bissau			Suitability level of Western zone		
Suitability level	Areas	Percentage	Very Low	1.23986E + 11	2.387602
Very Low		0.00	Low	2.96893E + 12	57.17292
Low	4396.75	14.23	Moderate	1.80601E + 12	34.77845
Moderate	23,464.02	75.95	High	2.79817E + 11	5.388466
High	3033.59	9.82	Very High	14154233738	0.27257
Very High		0.00		5.19289E + 12	100
	30,894.36	100.00			

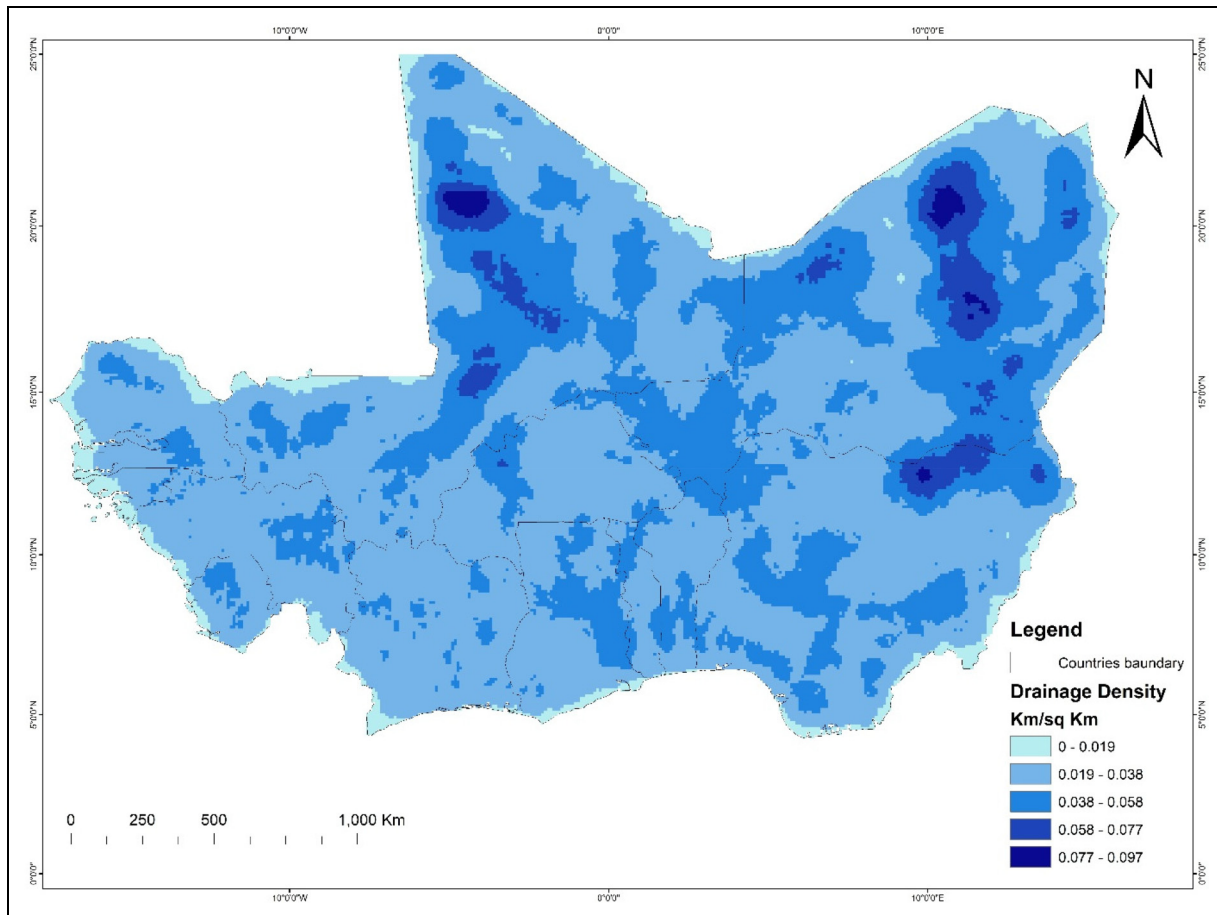


Figure 3. Map of Western drainage density.

sandy loam. Loam and sandy clay loam are the dominant textures in most countries, for example, Mali (54.88% loam, 20.94% sandy clay loam), Niger (40.42% loam, 19.96% loamy sand), and Benin (72.32% sandy clay loam). Silty loam is present only in Mali, while unweathered bedrock occurs in Mali and Niger. Figure 10 illustrates the spatial distribution of soil textures across the region.

Table 13 shows SPI values predominantly in the lowest category ($0.0-2.18 \times 10^9$), with near-total coverage in all countries: Mali (99.99%), Niger (99.97%), Burkina Faso (100%), Benin (99.99%), Nigeria (99.94%), Togo (100%), Ghana (99.97%), Côte d'Ivoire (100%), Guinea (100%), Senegal (100%), Sierra Leone (100%), The Gambia (100%), and Guinea Bissau (100%). Minor quantities appeared in higher SPI categories in Mali, Niger, Benin, Nigeria, and Ghana. Figure 11 presents the spatial variability of SPI, highlighting the generally low erosive power of streams in the WAPP.

Table 4 presents the spatial distribution of hydroelectric site suitability levels across the WAPP member countries, classified into five categories: Very Low, Low, Moderate, High, and Very High suitability. The areas (in km^2) and corresponding percentages indicate the extent of each suitability class within each country. The results show that

most countries have most of their land area falling within Low to Moderate suitability classes, with smaller proportions classified as High or Very High suitability. For instance, Mali and Niger predominantly exhibit Low suitability (71.38% and 66.13%, respectively), while Sierra Leone shows a considerable percentage (49.37%) in the High suitability category. No country reported significant areas under Very High suitability, suggesting that optimal hydroelectric sites are limited and dispersed. This classification supports targeted decision-making for hydroelectric development by highlighting priority zones within each country. Figure 11 represents the suitability map for west.

Interpretation. The geospatial multi-criteria assessment of hydroelectric site suitability across the West African Power Pool (WAPP) integrates nine biophysical parameters into a Composite Appropriateness Index (CAI). The analysis reveals distinct spatial patterns and quantitative distributions that influence hydroelectric potential.

Drainage density. Drainage density across WAPP ranges from 0.0 to 0.097 km/km^2 , with a predominant concentration below 0.046 km/km^2 in most countries. For example,

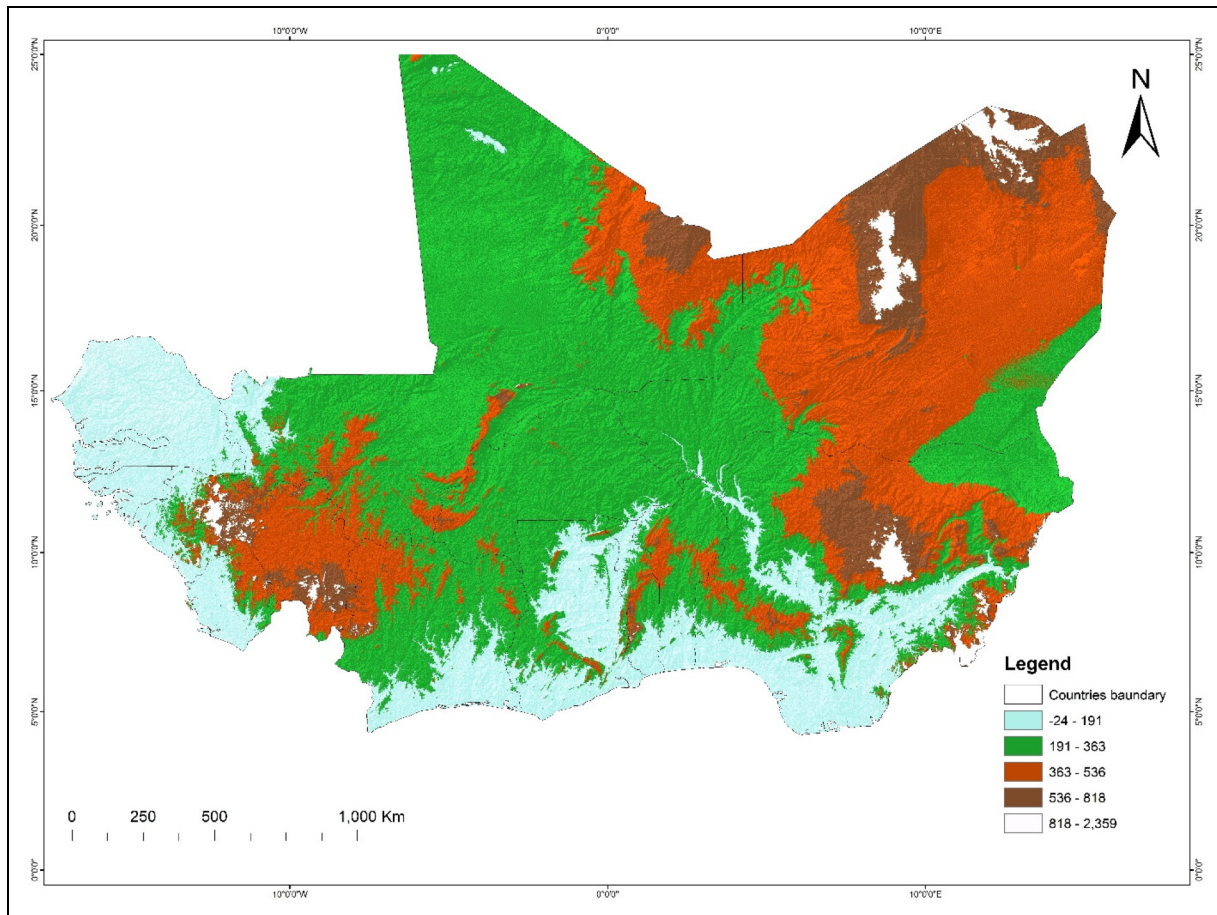


Figure 4. Map of West elevation.

Mali (79.60%), Niger (67.77%), and Burkina Faso (95.19%) exhibit low drainage densities, indicating dispersed runoff conducive to small-scale hydropower development.

Elevation. Elevations are largely below 536 m, covering 97.12% of Mali and 100% of Senegal, reflecting predominantly low-lying terrain. This elevation range suggests limited potential energy head for large-scale hydropower but suitability for low-head technologies.

Flow accumulation. Flow accumulation values are mostly low, consistent with a dendritic drainage pattern typical of the region, except in The Gambia where 60.55% of the area shows high flow accumulation, indicating localized potential for hydropower development.

Geology. Precambrian (undivided) formations dominate the geology in most countries, except The Gambia and Guinea Bissau, where Precambrian geology accounts for only 4.12%. Geological stability inferred from these formations supports infrastructure development feasibility.

Land cover. Bare ground and rangeland dominate land cover, e.g., Mali with 37.84% bare ground and 58.25% rangeland. Tree cover is more prevalent in southern and coastal countries, which may influence environmental constraints and site selection.

Rainfall. Annual rainfall varies significantly, from 1.89 mm to 4445.97 mm. Coastal countries such as Sierra Leone receive over 1430.97 mm (100%), aligning with higher hydroelectric suitability due to greater water availability.

Slope. Slope gradients are generally low (0.0° – 1.79°), covering over 80% of many countries (e.g., Mali 97.73%). Such low slopes limit potential energy head but favor low-head hydropower technologies.

Soil texture. Soil textures are predominantly loamy with clay and sand components, e.g., Mali with 54.88% loam and 20.94% sandy clay loam. These soil types affect construction feasibility and erosion risks.

Stream power Index (SPI). SPI values are mostly low (0.0 – 2.18×10^9), with near-total coverage in all countries (e.g.,

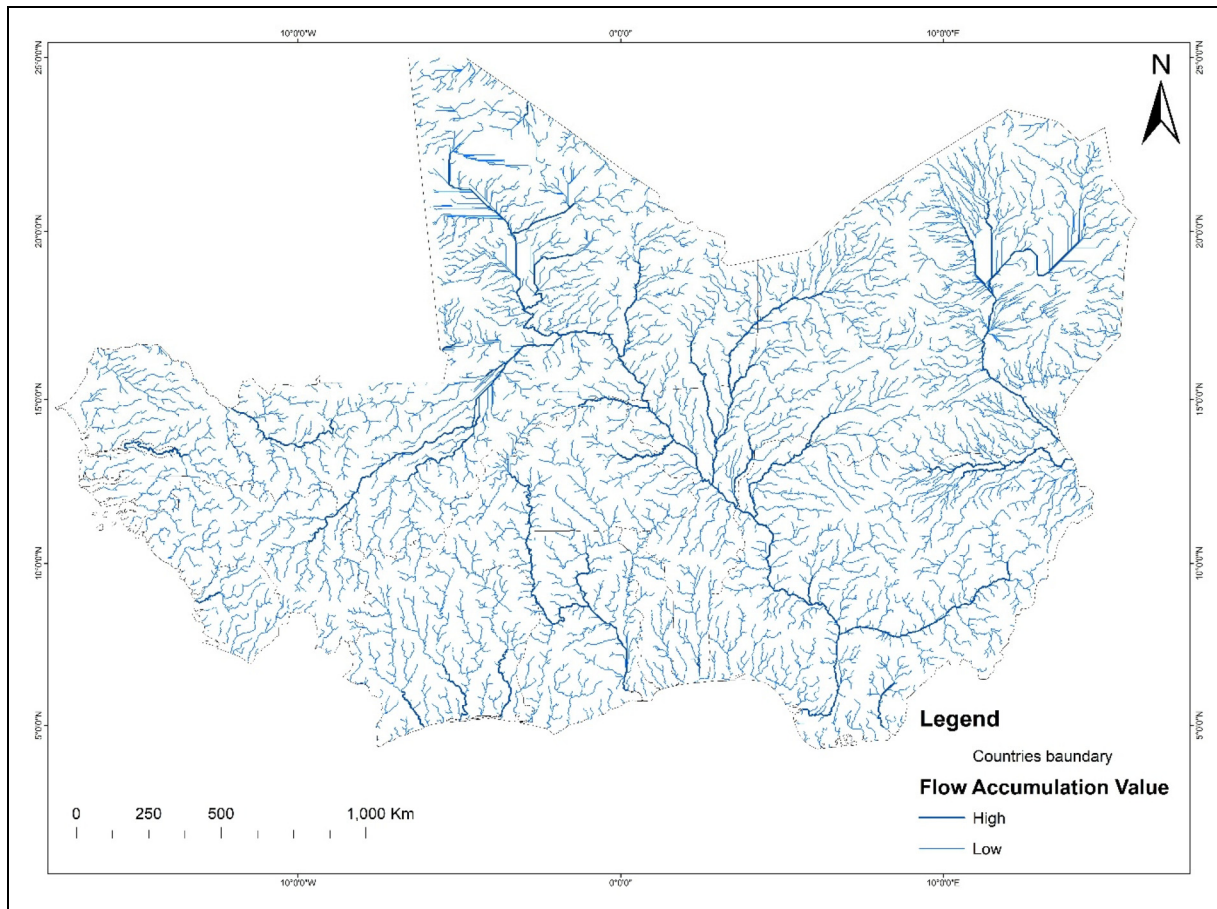


Figure 5. Map of West flow accumulation.

Mali 99.99%, Nigeria 99.94%), indicating generally low erosive energy of streams.

Composite appropriateness Index (CAI) suitability. Suitability classification shows most land areas fall under low to moderate suitability, e.g., Mali with 71.38% low and 23.83% moderate suitability. Sierra Leone stands out with 49.37% high suitability, highlighting priority zones for hydroelectric development. No areas were classified as very high suitability.

Validation. Spatial overlay with existing hydropower infrastructure, including Manantali (Mali), Akosombo (Ghana), and Kainji (Nigeria) dams, reveals strong correspondence with CAI high and moderate suitability zones, confirming the model's robustness.

Discussion

The results provide a comprehensive spatial characterization of hydroelectric site suitability in WAPP, revealing critical insights into regional hydropower potential and development feasibility.

Hydropower typology and feasibility

The findings indicate a strong regional bias toward specific hydropower technologies. The predominance of slope gradients between 0.0° and 1.79° covering over 80% of the territory in countries like Mali, Burkina Faso, and Senegal effectively precludes the development of traditional high-head storage schemes which require significant topographical relief for large-scale potential energy generation (NASA, 2025). Consequently, these regions are highly suited for low-head technologies and run-of-river micro/mini-hydropower systems. These systems can leverage the high flow accumulation found in localized basins (such as 60.55% of The Gambia) without requiring large-scale flooding or dam infrastructure. Conversely, the "High Suitability" zones in Sierra Leone and Nigeria, which combine higher rainfall with stable Precambrian geology, present the only viable candidates for large-scale storage schemes capable of providing baseload power to the WAPP grid.

Potential energy implications

Rainfall distribution strongly influences hydropower potential, with coastal regions exhibiting higher precipitation and

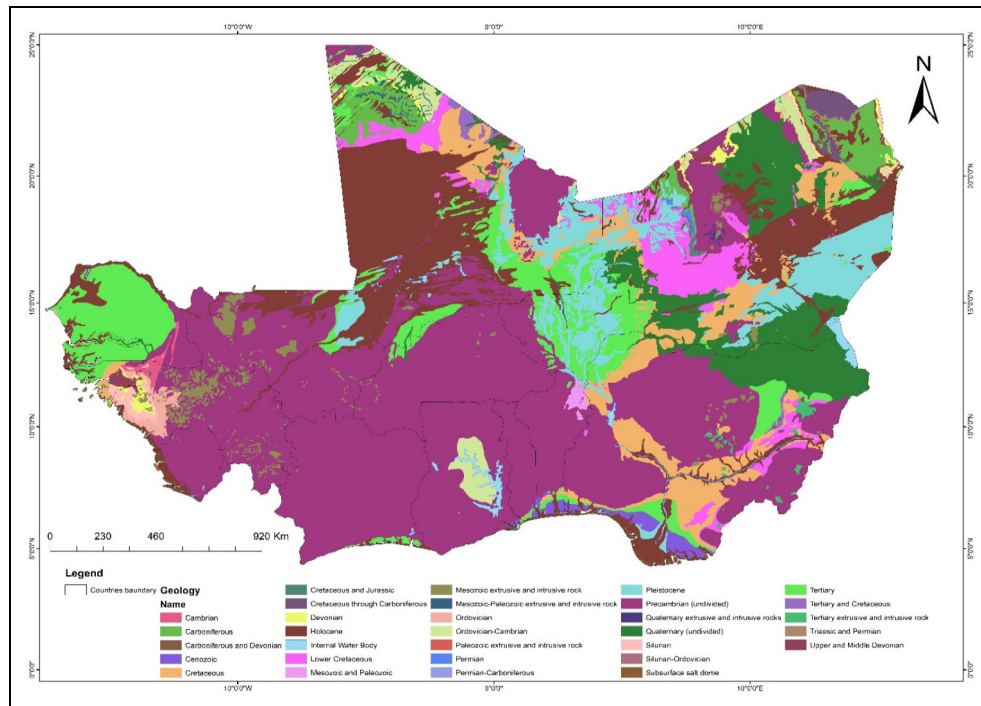


Figure 6. Map of West geology.

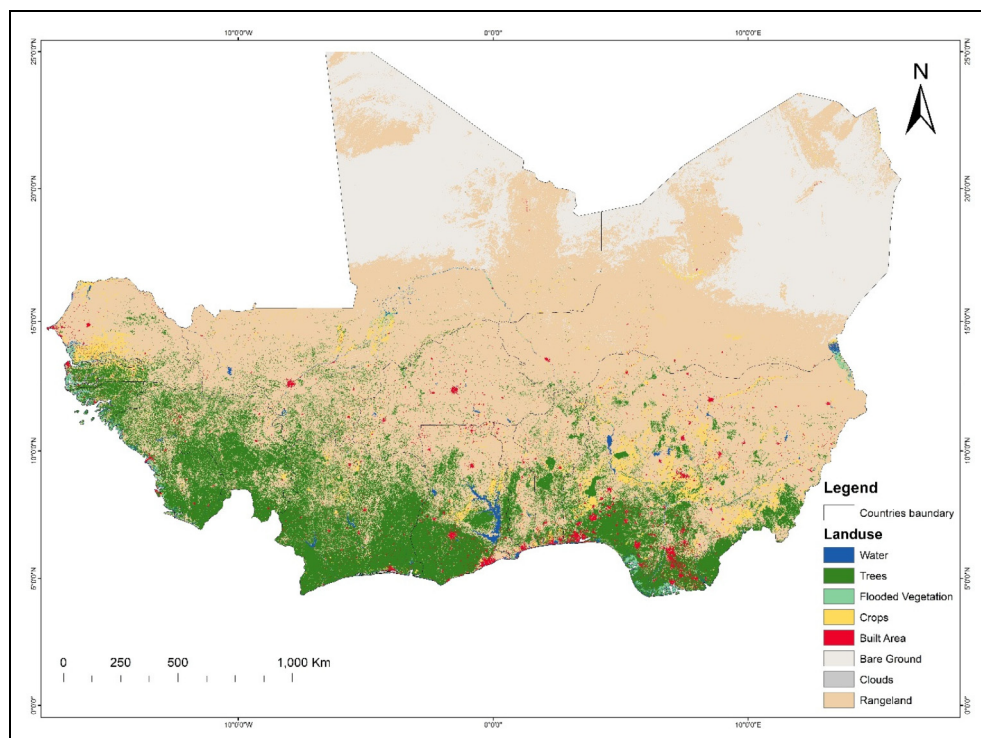


Figure 7. Map of West land cover.

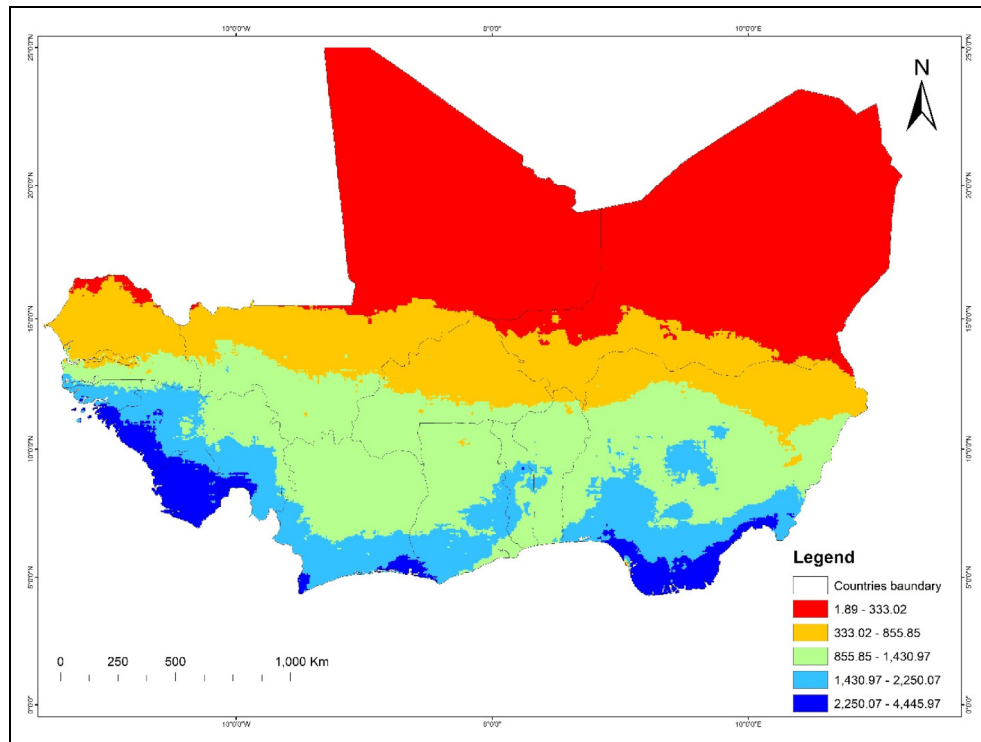


Figure 8. Map of Western rain fall.

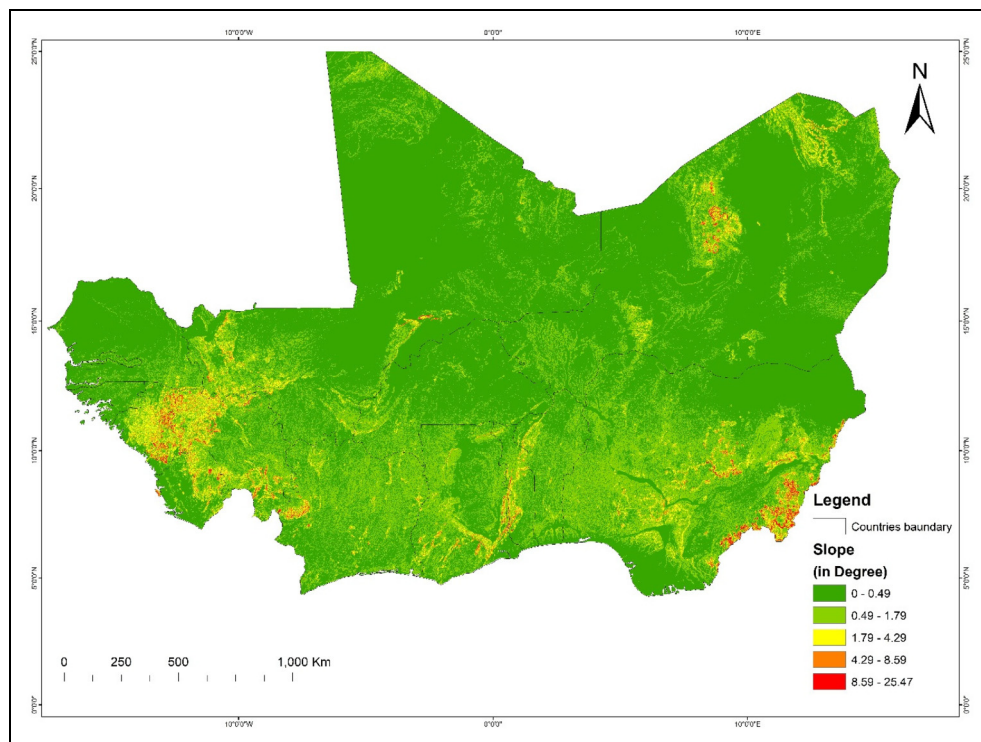


Figure 9. Map of Western slope.

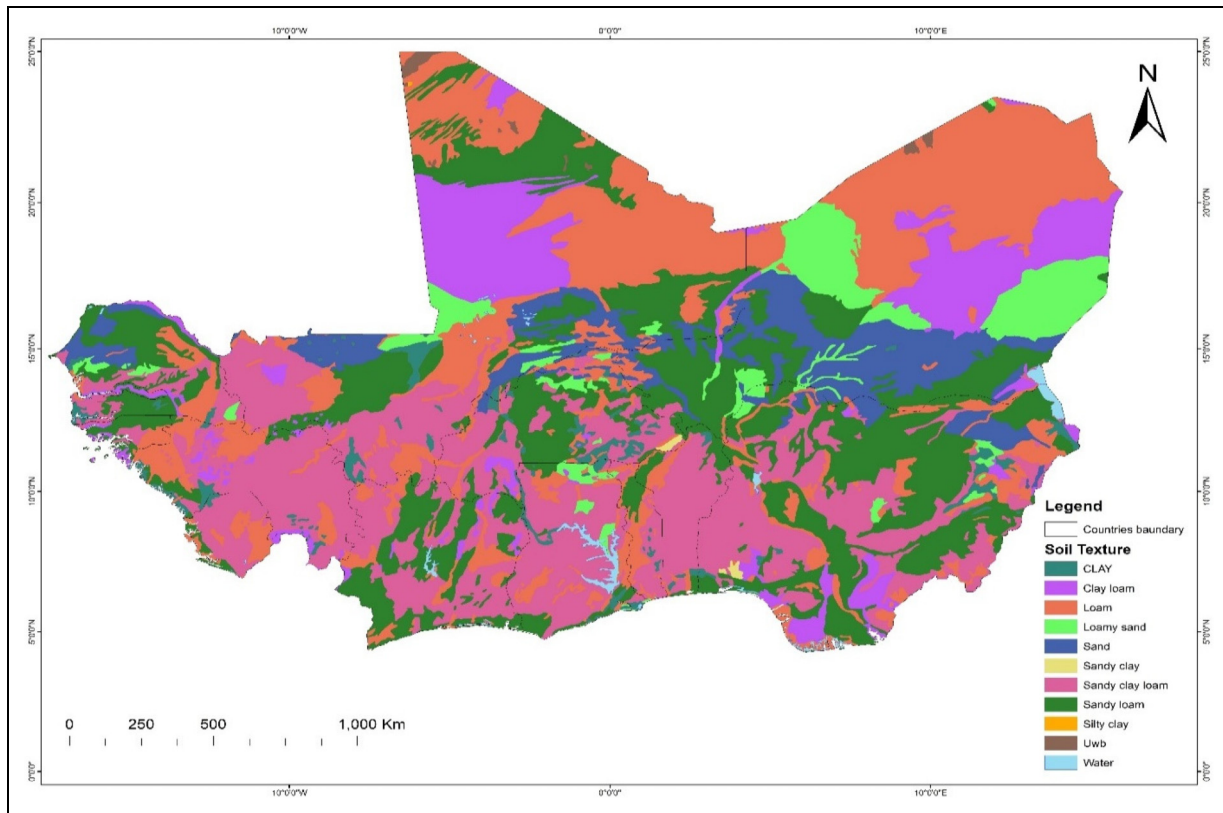


Figure 10. Map of Western soil texture.

corresponding higher CAI suitability scores. However, the generally low slope gradients limit the potential energy head, constraining large-scale hydropower capacity but favoring low-head technologies. The low SPI values imply limited stream erosive power, which may reduce sedimentation risks but also indicate moderate Stream Energy available for power generation.

Spatial prioritization utility

The CAI-based suitability map serves as a valuable decision-support tool, enabling policymakers and planners to prioritize hydropower development in zones with the most favorable biophysical conditions. The strong spatial alignment with existing hydropower infrastructure validates the model's practical relevance.

Limitations and future directions. While the static geospatial parameters provide a robust spatial framework, the absence of dynamic hydrological modeling (e.g., flow variability, discharge forecasting) limits the precision of energy potential estimates. Additionally, the current analysis does not incorporate socioeconomic and infrastructural factors such as grid proximity, population density, land use conflicts, and regulatory constraints, which are critical for practical project feasibility.

Conclusion

The drainage This study delivers a comprehensive geospatial multi-criteria assessment of hydroelectric site suitability across the West African Power Pool (WAPP) by integrating nine critical biophysical parameters into a Composite Appropriateness Index (CAI). The findings highlight that the region predominantly exhibits low to moderate drainage densities ($0.0\text{--}0.097\text{ km/km}^2$), with most countries such as Mali (79.60%), Niger (67.77%), and Burkina Faso (95.19%) falling below 0.046 km/km^2 . Elevations are largely under 536 m, covering 97.12% of Mali and 100% of Senegal, whereas flow accumulation patterns are primarily dendritic and low, except for The Gambia, with 60.55% high flow accumulation. Geological formations are mainly Precambrian (undivided), except in The Gambia and Guinea-Bissau (4.12%). Land cover is dominated by bare ground and rangeland, exemplified by Mali's 37.84% bare ground and 58.25% rangeland coverage. Rainfall distribution shows significant spatial variability, with coastal countries such as Sierra Leone receiving over 1430.97 mm annually (100%). Slope gradients are generally low ($0.0^\circ\text{--}1.79^\circ$), covering over 80% of many countries (e.g., Mali 97.73%), indicating a limited potential energy head for large-scale hydropower but favorable conditions for low-head technologies. Soil textures are predominantly loamy with clay and sand components, such as Mali's 54.88% loam and 20.94%

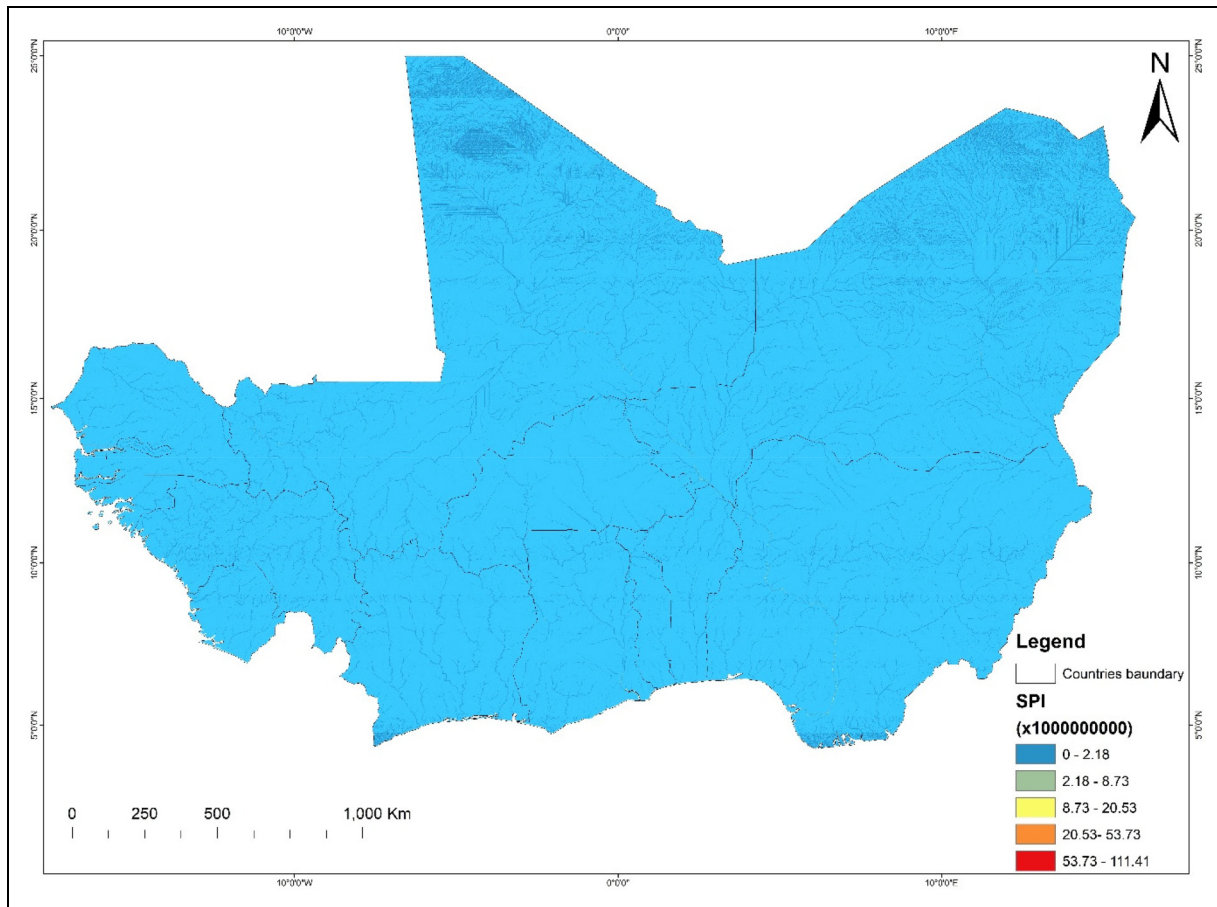


Figure 11. Map of Western SPI (stream power Index).

sandy clay loam content. The Stream Power Index values were mostly low ($0.0\text{--}2.18 \times 10^9$), with near-total coverage in all countries.

The CAI suitability classification revealed that most land areas fall under the low to moderate suitability categories, with exceptions such as Sierra Leone, which exhibited 49.37% high suitability. Validation against existing hydro-power infrastructure, including the Manantali, Akosombo, and Kainji dams, demonstrated a strong spatial correspondence, confirming the model's robustness as a spatial decision-support tool.

This study underscores the critical role of geospatial multi-criteria analysis in guiding regional hydroelectric planning in the WAPP. To advance towards a comprehensive and energy-focused assessment, to refine these geospatial findings, future work must transition from static suitability mapping to dynamic energy estimation. This should include: Hydrological Modeling: Integration of models such as SWAT or HBV to estimate specific annual energy potential (GWh/year) by accounting for seasonal flow variability, Socioeconomic Constraints: Inclusion of 'exclusion zones' such as protected areas, high population density centers, and land-use conflicts (Esri, Land Use

and Land Cover, 2020a; Esri, 2020b), and Infrastructural Proximity: Evaluating site viability based on the distance to the existing WAPP regional grid to account for transmission costs and technical feasibility. Such multidimensional approaches will enhance the practical feasibility and sustainability of hydroelectric development, thereby supporting West Africa's renewable energy goals and regional energy security.

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Author contributions

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All authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as potential conflicts of interest.

Data availability statement

The data for this study is available from the corresponding authors upon request.

Institutional review board statement

The authors declare that the content of this study complies with ethical standards and concepts. The authors confirm that this paper has not been published previously, it is not under consideration for publication elsewhere and is not under consideration for publication elsewhere.

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